PRIME SCHOOL

Humanities

Year 9

Cambridge Lower Secondary
Student Manual

Humanities

Year 9

Cambridge Lower Secondary

Student Book · Prime Books

ABOUT THIS BOOK

One subject, in two parts. Part one builds your position on the planet: the lines, the coordinates, the climates and the two ways of halving a world. Part two then runs one unbroken causal line from the Renaissance to the world you live in.
Ten chapters, 60 numbered topics, and eight Portugal in the global story sections, with the answers, a glossary, the sources and an index at the back.

Prime Books is the student book series of Prime School.

First edition, 2026.

Written in British English. Distances are metric, money is in euros and times are given on the 24 hour clock.

This book was written for the pupils of Prime School by the Prime School Humanities Department. Every fact, date and figure in it was checked against a named source before it was printed, and those sources are listed at the back of this book, where you can check them too.

EDITORIAL BOARD

The Prime Books Subject Series has been developed under the guidance of the Pedagogical Academic Group for each subject and coordinated by the Pedagogical Team of Prime School.

This publication reflects our shared commitment to academic excellence, educational quality, professional integrity, and the continuous advancement of teaching and learning. Through the collective expertise and dedication of our educators, we aim to provide meaningful and engaging learning resources that support students' academic growth.

We extend our sincere appreciation to the Pedagogical Department and the Content Creation Team of Prime School for their invaluable contributions, dedication, and collaborative efforts in making this publication possible.

HOW TO CHECK THIS BOOK

Everything in these pages can be checked, and the book is arranged so that you can.

√ Every **figure, map, graph and table** carries a number, a title and a caption saying what to look for. A figure numbered 7.2 is the second figure of chapter 7.

√ Every **date, figure and statistic** was verified against a source before it was printed, and every source is named in the Sources section at the back.

√ Every **square code** leads to a checked English-language resource from a museum, an archive, a national survey or a public database. No code appears twice, and the full web address is printed beside each one, so the page still works without a device.

√ Every **closed question** has an answer at the back. Open questions have acceptance criteria instead: what a good answer must contain, rather than the words it must use.

If you find something in this book that is wrong, tell your teacher. That is how the next edition gets better, and it is also the habit this subject is trying to teach you.

Introduction: why study Humanities?

This book is about two questions that look separate and are not: where things are, and how they came to be that way. Geography answers the first. History answers the second. Neither answer is complete without the other.

Geography helps you understand where places are, why environments differ, how climate shapes the way people live, why populations are spread as unevenly as they are, how natural resources shape economies, and why location matters.

History helps you understand how societies change, why industry began where it did, why new political ideas appeared, why nations went to war, and how the international system you live in was built.

The two are joined, and the joins are the interesting part. Location, climate, resources, oceans, borders and distance shaped what happened. War, industrialisation, migration, technology and political decisions then reshaped territories, cities and populations. So the course has one central idea, and it runs through every chapter:

> **THE IDEA THIS BOOK IS BUILT ON**
>
> **Place shapes people. People reshape place.**
>
> Or, set out as this book teaches it:
>
> place, environment, resources, people, technology, economy, ideas, power, conflict, change, connection.

Where this book sits in your Humanities course

Year 9 completes a journey you began two years ago.

In **Year 7** you asked: *how do civilisations begin?* You went from the formation of the Earth to continents and oceans, to human settlement, agriculture, cities and the first civilisations.

In **Year 8** you asked: *how do states, empires and networks develop?* From Greek city-states and Alexander, through the Roman republic and empire, to medieval kingdoms, trade networks, stronger monarchies, the Renaissance and maritime expansion.

This year you ask the question those two were leading to:

> **THE YEAR 9 QUESTION**
>
> **How was the modern world created?**

The line this book follows

Part one puts you on the planet. Everything after it is one causal chain, and no link in it is skipped:

Industrial Revolution → technological innovation → industrial capitalism → new political ideologies → nationalism and imperial rivalry → **the First World War** → interwar instability → **the Second World War** → a divided world → **the Cold War** → the nuclear and space races → the end of the Cold War → the modern connected world.

That chain is the reason the book does not jump from the Industrial Revolution to

1 Parts three and four exist so that when you reach the First World War you

can explain **why** it happened, not only when.

Portugal in the global story

At the end of every historical chapter with a Portuguese dimension there is a section called **Portugal in the global story**. It always answers the same four questions:

- √ What was happening in Portugal?
- √ What was happening in the world at the same time?
- √ What part did geography play?
- √ Was Portugal participating, resisting, following later, developing

differently, or influencing events elsewhere?

Portugal is never a separate national history bolted on at the end. It is the same chronology, seen from here.

Contents

Every unit and every numbered topic in this book, with the page it really starts on.

The shape of a chapter

Every chapter opens with a full page that carries five things: an illustration, the chapter number and title, **the question the chapter answers**, what you will be able to do by the end, and the list of topics in the order you meet them. That page also carries the chapter's **square code**.

Then the topics run in sequence. Each one begins with its own label, giving the chapter and the topic number, so you always know where you are.

The panels, and what each is for

PANEL	WHAT IT IS FOR
Before you read	One question from ordinary life, before any teaching
The account	The teaching itself, in continuous prose
Worked enquiry	One question answered slowly, showing the reasoning
Your turn	Numbered practice. Closed questions have answers at the back
Go further	The stretch, for when you want more
From the archive	A verified, sourced fact worth knowing
On the map	The geography inside a history topic, and the reverse
Read the source	A source, with origin, purpose, value and limitation
Careful	The common mistake, named before you make it
The enquiry	The chapter's big question, to be argued rather than answered
What you can now do	Your own check, at the end of every chapter

Table 0.1 · The eleven panels.
Learn the first column. Every one of them appears in every part of the book, and they always do the same job.

The figures

Every map, diagram, graph, timeline and table carries a **number**, a **title** and a caption saying **what to look for**. A figure numbered 12.3 is the third figure of chapter 12. Nothing in a figure of this book was invented: every plotted number came from a source named at the back.

How to use the answers honestly

The answers to every closed question are at the back. Used well they are the most useful pages in the book; used badly they are the most useless.

- ✓ Attempt the question first, in writing, even if you are unsure.
- ✓ Then check, and if you were wrong, find out **why** before you move on.
- ✓ For open questions there is no single answer, so the back of the book gives

acceptance criteria instead: what a good answer has to contain.

The square codes

Each chapter carries one code, and it leads to a checked English-language resource from a museum, an archive, a national survey or a public database. No code appears twice in this book. The full address is printed beside each one, so the page works with no device in the room.

Ask your teacher before you scan. The codes are extension, never a substitute for the pages.

The two centuries at a glance

This is the whole book on one page: the world above, Portugal below. Come back to it whenever you lose your place in the chronology.

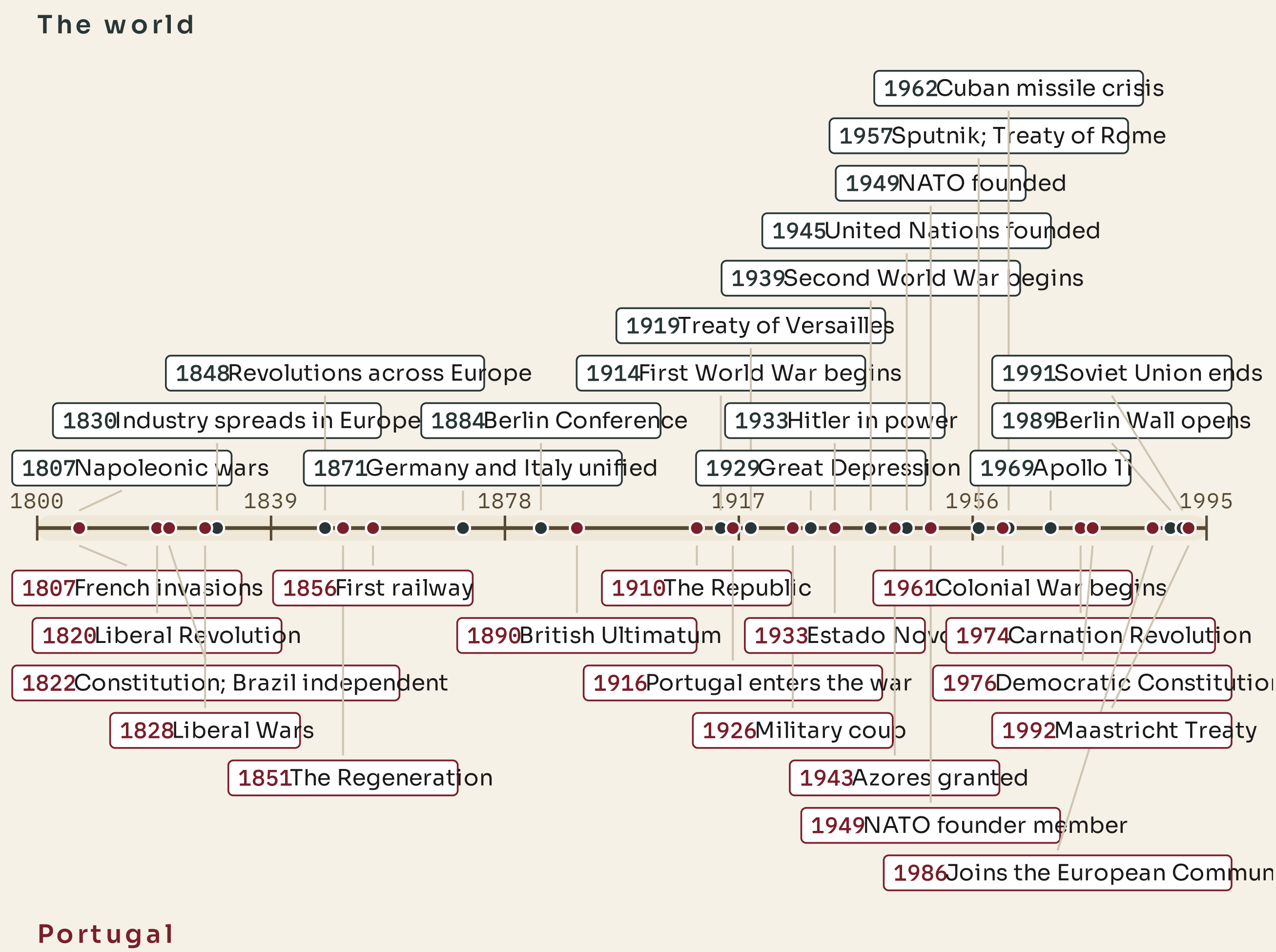


Figure 0.1 · Two centuries, two lines, one chronology.
Read the upper line, then the lower. Then read any year vertically, and ask what was happening in both places at once. You will meet this figure again on the last pages of the book, when you can explain every entry on it.

The skills this year is built on

Six things are being taught alongside the content, and they are what the questions in this book are really testing.

Chronology. Putting events in order, and knowing what was happening at the same time somewhere else.

Cause and consequence. Distinguishing a trigger from a cause, ranking causes, and saying what followed from what.

Source analysis. Asking of any source: who made it, for whom, why, what it shows, and what it had reason **not** to say.

Interpretation. Recognising that historians disagree, and that disagreement is usually about weight rather than about facts.

Map and data skills. Reading a map, a graph and a table, and saying what each does and does not show.

Geographical explanation. Explaining why something happened **where** it happened.

> **THE HABIT THIS BOOK IS TRYING TO BUILD**
>
> When you meet any event in this book, ask two questions in this order. **Where did this happen, and why there?** Then: **what changed because of it?**
>
> If you can do that reliably by the last page, you have learned Humanities rather than a list of dates.

PART ONE

Understanding our global position

Everything in the rest of this book happens somewhere. Before you can explain why industry began where it did, why one country was fought over and another was left alone, or why a small state on the edge of Europe mattered to two world wars, you need to be able to say exactly where anything is and what its position does to it. That is what these four chapters are for.

PART ONE · CHAPTER 1

The global grid

Five lines nobody can see, and what each one decides

THE QUESTION THIS CHAPTER ANSWERS

How do a few imaginary lines let us say exactly where anything on Earth is?

IN THIS CHAPTER, YOU WILL

- √ Say where the Equator, the Tropics and the polar circles are, and why.
- √ Explain why equatorial places are hot all year and polar places are not.
- √ Name the continents the Equator crosses, from a map and then from memory.
- √ Tell the Arctic and Antarctica apart, and say which is ocean and which land.
- √ Explain what a meridian is and why one of them was chosen as zero.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Encyclopaedia Britannica, the Equator. The line this chapter is built on, and why its position decides how much solar energy the ground below it receives.

www.britannica.com/science/Equator

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Somebody sends you a message saying only: "Meet me at the beach." In pairs, write down every question you would have to ask before you could actually go. Now do the same for "Meet me at fifteen degrees north, sixty-one degrees west." Which instruction is more useful, and why? Keep your answers. You will come back to them at the end of chapter 2.

CHAPTER 1 · TOPIC 1.1

The Equator

The account

The Equator is a line drawn exactly half way between the North Pole and the South Pole. Its position is not a matter of opinion or agreement: it is the one line the shape of the Earth decides for us. Everywhere on it is the same distance from both poles, and it divides the planet into a northern half and a southern half. Its latitude is zero degrees, and every other latitude on Earth is measured from it.

You cannot see it. There is no mark on the ground, no rope across the ocean, nothing painted on the sea. Yet an enormous amount follows from where it is.

The reason is sunlight. Because the Earth is a ball, the Sun's rays do not strike every part of the surface at the same angle. Near the Equator they arrive close to straight down, so a beam of a given width lands on a small patch of ground and heats it strongly. Near the poles the same beam arrives at a shallow angle and is smeared across a much larger patch, so it heats the ground far less. Nobody near the Equator has hotter sunshine. They have more concentrated sunshine.

Two consequences follow, and both matter for the rest of this book. First, equatorial places are warm all year rather than warm in one season, because their angle to the Sun barely changes. Second, warm air rises, and rising air cools and drops its moisture as rain. That is why a belt of very heavy rainfall, and the rainforests that live on it, sits along the Equator rather than anywhere else.

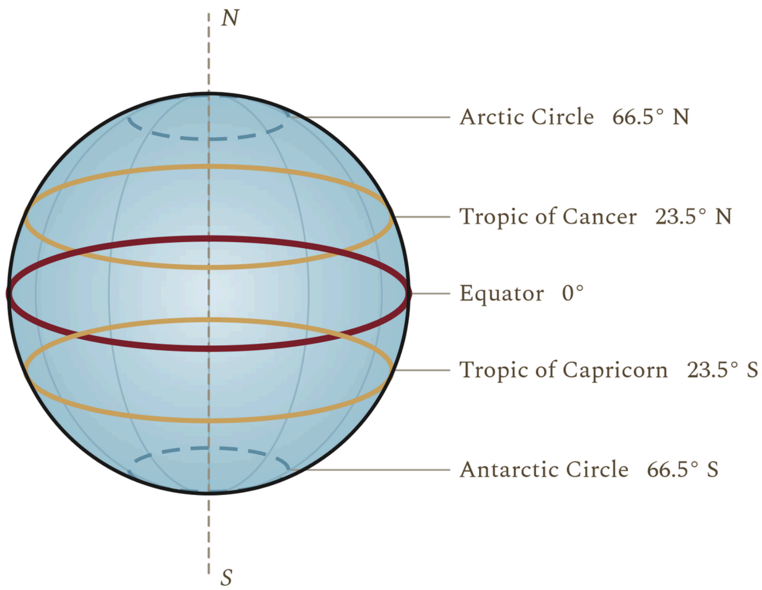


Figure 1.1 · The five named lines, and the angles that put them there.
Follow the Equator round first. Notice that it is the only one of the five that is a complete circle at full size, and that the four others are placed by the tilt of the Earth's axis rather than chosen.

The Equator crosses three continents: South America, Africa and Asia, if the islands of Indonesia are counted as Asia. It also crosses a great deal of ocean. Thirteen countries have territory on it, and the list is worth looking at because it is a list of places whose climate, farming and history are all shaped by their position.

CAREFUL

The Equator is the hottest **belt** on average, but it is not where the highest air temperature has ever been measured. The record heat is found in the desert belts near the Tropics, where there is no cloud to shade the ground and no daily rain to cool it. Concentrated sunlight and extreme heat are not quite the same thing, and the difference is explained by rainfall.

YOUR TURN

1. In one sentence, say what the Equator is, without using the word "middle".
2. Explain why sunlight is more concentrated at the Equator than at the poles.
   Draw the two beams if it helps.
3. Name the three continents the Equator crosses.
4. A pupil writes: "It is hot at the Equator because it is closer to the Sun."
   The distance from Earth to the Sun is about 150 million kilometres, and the Earth's radius is about 6 400 kilometres. Use those two figures to explain why the pupil is wrong.

CHAPTER 1 · TOPIC 1.2

The Tropics

The account

The Earth spins on an axis, and that axis is tilted. It leans at about twenty-three and a half degrees away from upright, and, crucially, it keeps leaning the same way all year as the Earth travels round the Sun. Nothing about the tilt changes. What changes is which half of the planet happens to be leaning toward the Sun at that point in the orbit.

That single fact produces the seasons, and it also produces two more named lines.

The Tropic of Cancer sits at about twenty-three and a half degrees north. It marks the furthest north that the Sun can ever be directly overhead. The Tropic of Capricorn sits at the same angle south, and marks the furthest south. Between those two lines, there is at least one day each year when the Sun is exactly overhead at noon and an upright post casts no shadow at all. Outside them, that never happens, anywhere, ever.

Notice that the number is not a coincidence. The Tropics are at twenty-three and a half degrees because the axis is tilted by twenty-three and a half degrees. The lines are a direct measurement of the lean.

Figure 1.2 · The tilt, and why it makes seasons.
The orbit is drawn round rather than stretched, because the seasons are not caused by the Earth being nearer the Sun at some times of year. Follow the axis: it points the same way at all four positions.

The belt between the Tropics is called the tropical zone, and it is the part of the Earth that receives the most solar energy over a year. It is not uniform. It contains rainforest, savanna and some of the driest desert on the planet, because rainfall varies within it even where temperature does not. Chapter 3 takes that apart properly.

ON THE MAP

Find the Tropic of Cancer on a world map and follow it east from the Atlantic. Note the countries it passes through in north Africa and Asia. Now do the same for the Tropic of Capricorn in the southern hemisphere. Write one sentence saying what the land along both lines has in common.

YOUR TURN

1. What is the tilt of the Earth's axis, and what does it stay doing all year?
2. Explain, in your own words, what the Tropic of Cancer marks.
3. Why is the Tropic of Capricorn at the same angle as the Tropic of Cancer?
4. In which of these places can the Sun be directly overhead at noon: Lisbon, Nairobi, Sydney, Singapore? Use the latitudes to decide, and say how you did it.

CHAPTER 1 · TOPIC 1.3

The polar circles

The account

The same tilt that puts the Tropics where they are also puts two further lines near the poles. The Arctic Circle lies at about sixty-six and a half degrees north; the Antarctic Circle at about sixty-six and a half degrees south. The number is simply ninety minus twenty-three and a half.

What these lines mark is not temperature but daylight. On or inside the Arctic Circle there is at least one day a year when the Sun does not set at all, and at least one when it does not rise. The further in you go, the more of those days there are, until at the pole itself the year is one very long day followed by one very long night. The same is true, in the opposite months, inside the Antarctic Circle.

People who live there call the summer version the midnight Sun. In northern Norway, Sweden, Finland and Russia, and across northern Canada, Alaska and Greenland, there are weeks in June when the Sun circles the sky without touching the horizon, and weeks in December of polar night, when it never clears it.

It is worth pausing on how strange this is if you have never seen it. The question "what time does it get dark?" has no answer in Tromso in June, and no answer in December either.

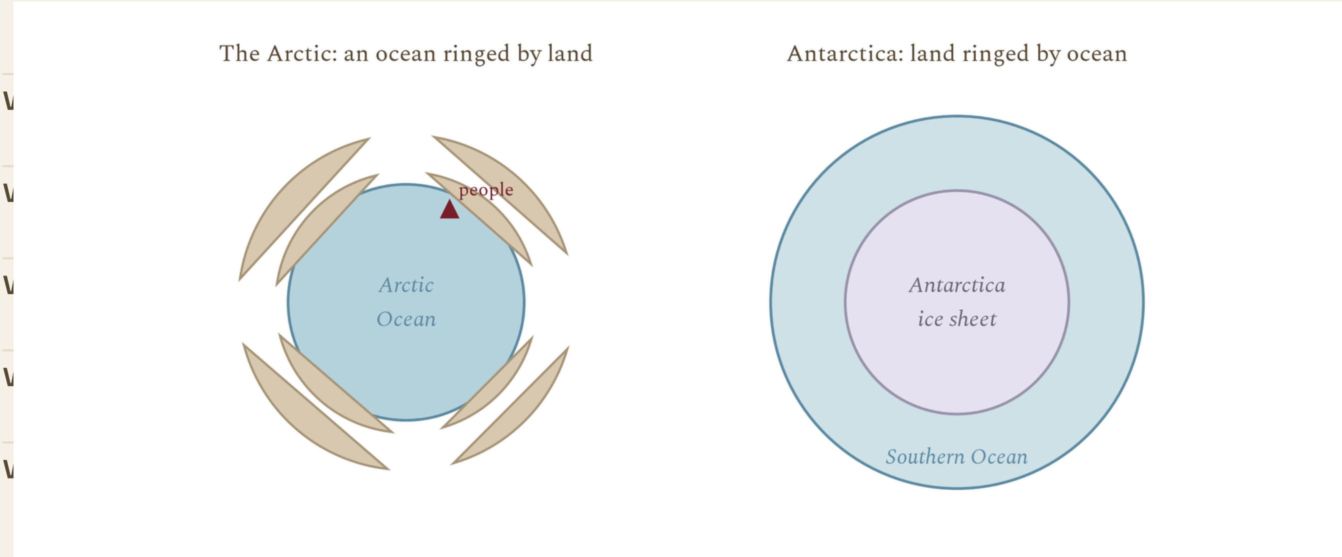


Figure 1.3 · The Arctic and Antarctica are not two versions of the same place.
Read the rows across rather than down. The difference in the first row explains almost every difference in the rows below it.

The two polar regions are often treated as a matching pair, and they are not. The Arctic is an ocean with sea ice floating on it, ringed by the northern edges of three continents, and people have lived around it for thousands of years. Antarctica is a continent of rock buried under an ice sheet kilometres thick, ringed by ocean, with no permanent population at all. That difference is geography deciding history: one is a place with nations, towns and disputes over shipping routes, the other is governed by a treaty because nobody lives there to govern.

YOUR TURN

1. At what latitude does each polar circle sit, and where does that number come from?
2. Explain the midnight Sun to somebody who has never left Portugal.
3. Give two differences between the Arctic and Antarctica.
4. Why does Antarctica have no permanent population when the Arctic has had one for thousands of years? Give a geographical reason, not just a cold one.

CHAPTER 1 · TOPIC 1.4

Meridians

The account

Latitude alone cannot find a place. If you know only that somewhere is at forty degrees north, you know it sits on a circle running right round the planet, and that circle passes through Portugal, Spain, Italy, Greece, Turkey, China, Japan and the United States. You need a second measurement, running the other way.

That measurement is longitude, and the lines that carry it are called meridians. A meridian runs from the North Pole to the South Pole, crossing every parallel of latitude at a right angle. Every meridian is the same length as every other meridian, because each is half of a circle round the whole Earth. Parallels are not like that at all: the Equator is a full-sized circle, but a parallel near the pole is a small one.

Here is the difficulty. Latitude has a natural zero, because the Equator is fixed by the shape of the planet. Longitude has no natural zero whatsoever. One meridian is exactly as good as another. Somebody has to choose.

For centuries, different countries chose differently, and each printed charts measured from its own capital. A French chart measured from Paris, a Spanish chart from Cadiz, a Portuguese chart from Lisbon. A navigator carrying charts from two countries was carrying two incompatible grids, which is precisely the kind of problem that sinks ships.

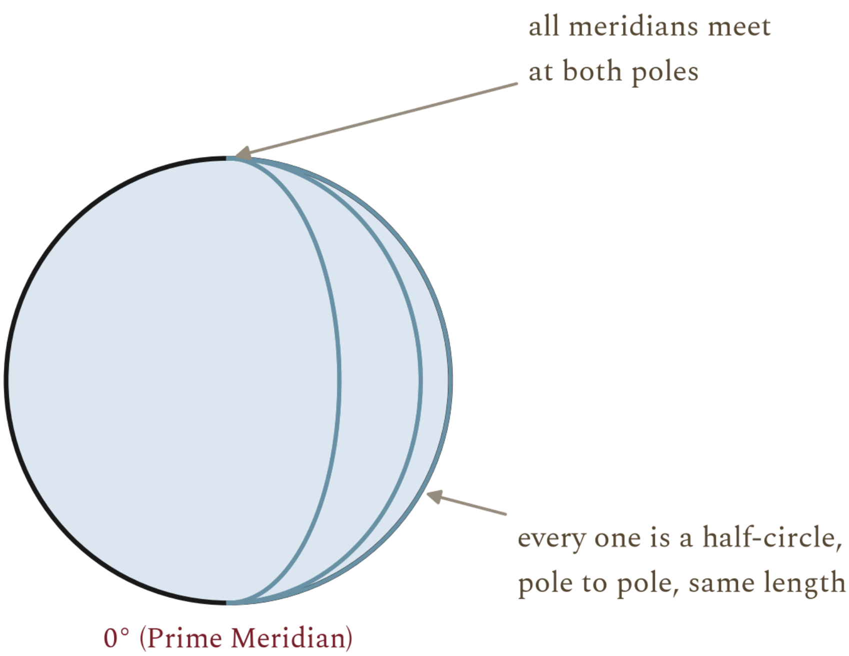


Figure 1.4 · Meridians all run pole to pole, and all are the same length.
Compare them with the parallels in the first figure of this chapter. Notice that meridians converge: they are furthest apart at the Equator and meet at the poles.

FROM THE ARCHIVE

In October 1884, delegates from twenty-five countries met in Washington at the International Meridian Conference and voted to adopt a single prime meridian for the world. Greenwich won largely because most of the world's shipping was already using charts measured from it. The vote was twenty-two in favour, one against, and two abstentions. France abstained and went on using Paris in its own charts for decades.

YOUR TURN

1. What is a meridian, and how is it different from a parallel?
2. Explain why latitude has a natural zero and longitude does not.
3. Why was it a practical problem for navigation that countries used different prime meridians?
4. Suggest one argument that a French delegate in 1884 might have made for Paris. Then say why it lost.

CHAPTER 1 · TOPIC 1.5

The Prime Meridian

The account

The line the 1884 conference chose runs through the Royal Observatory at Greenwich, in London, and it is called the Prime Meridian. Its longitude is zero degrees. Everything east of it is measured in degrees east, up to one hundred and eighty; everything west of it in degrees west, up to the same figure. The two meet on the far side of the planet at the one hundred and eightieth meridian, which is a single line that can be called either.

Choosing a prime meridian solved three problems at once.

It solved **navigation**, because every chart could now be drawn on the same grid, and a position reported by one ship meant the same thing to another.

It solved **mapping**, because a survey made in one country could be joined to a survey made in the next without a correction being applied by hand.

It solved **time**, and this is the part that reaches furthest into everyday life. Once the world agreed on a zero meridian, it could agree on a zero for clocks. Chapter 2 follows that through, but the link is worth seeing now: the reason your phone knows what time it is in Tokyo runs back to a decision about where to draw a line.

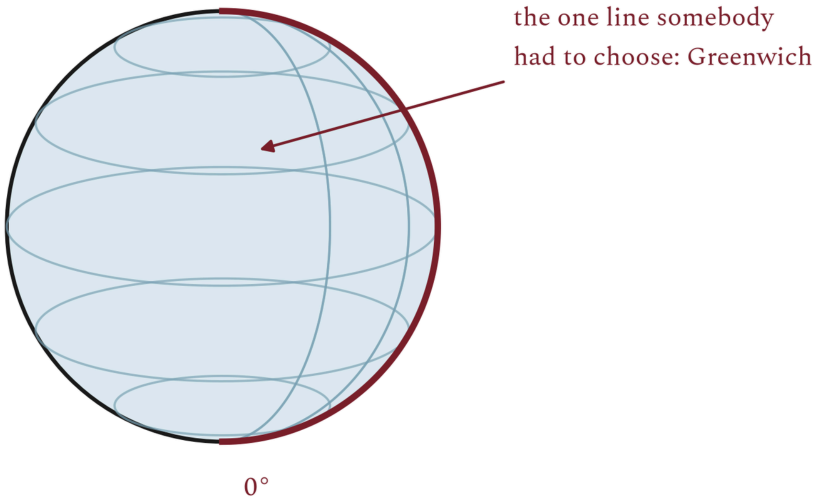


Figure 1.5 · The grid, and the one line somebody had to choose.
Follow the Prime Meridian from Greenwich down through western Europe, Africa and the Atlantic. Note how much of the map lies east of it, and how much west, and that this is an accident of the choice rather than a fact about the Earth.

Portugal sits just west of the Prime Meridian. Lisbon is at about nine degrees west, so mainland Portugal lies wholly in the western hemisphere as geographers define it, although it is in western Europe politically and culturally. Chapter 4 comes back to why those two statements are not the same statement.

WHAT YOU CAN NOW DO

- √ Say where the Equator is and explain why it decides so much about climate.
- √ Explain why the Tropics and the polar circles sit where they do.
- √ Describe the midnight Sun and the polar night, and say who experiences them.
- √ Give two real differences between the Arctic and Antarctica.
- √ Explain what a meridian is and why the world had to choose one as zero.

WORDS FOR THIS CHAPTER

Equator, latitude, longitude, meridian, parallel, axis, tilt, Tropic of Cancer, Tropic of Capricorn, Arctic Circle, Antarctic Circle, midnight Sun, polar night, Prime Meridian, hemisphere

CHAPTER 1 AT A GLANCE

The global grid

how do a few imaginary lines let us say exactly where anything on Earth is?

THE TOPICS YOU HAVE COVERED

1.1	The Equator
1.2	The Tropics
1.3	The polar circles
1.4	Meridians
1.5	The Prime Meridian

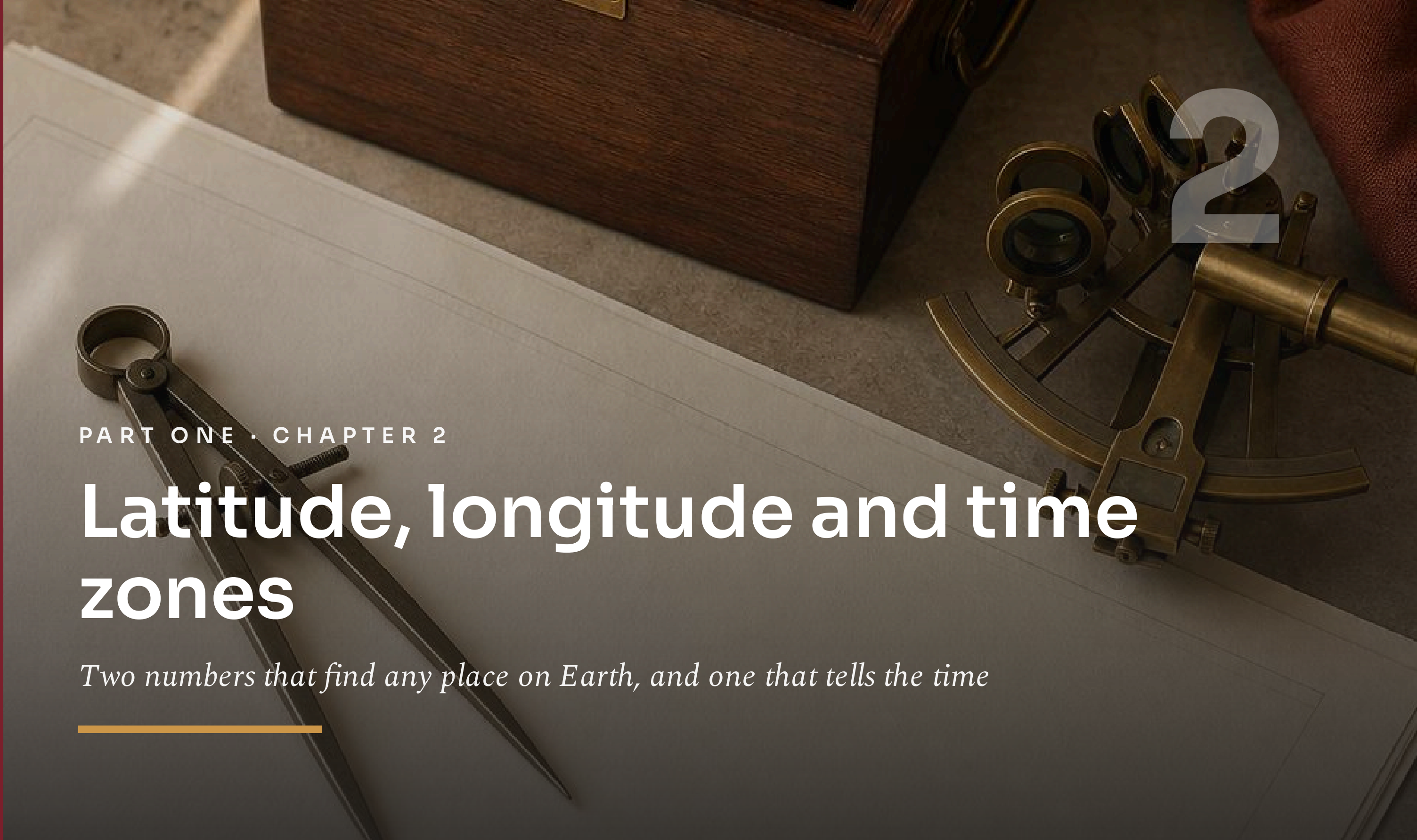
WHAT YOU CAN NOW DO

- √ Say where the Equator is and explain why it decides so much about climate.
- √ Explain why the Tropics and the polar circles sit where they do.
- √ Describe the midnight Sun and the polar night, and say who experiences them.
- √ Give two real differences between the Arctic and Antarctica.
- √ Explain what a meridian is and why the world had to choose one as zero.

THE WORDS THIS UNIT TAUGHT

Equator · latitude · longitude · meridian · parallel · axis · tilt · Tropic of Cancer · Tropic of Capricorn · Arctic Circle · Antarctic Circle · midnight Sun · polar night · Prime Meridian · hemisphere

Use this page to revise before you go on. If you cannot do one of the things on the right, go back to the topic that taught it, not to the answers.



2

PART ONE · CHAPTER 2

Latitude, longitude and time zones

Two numbers that find any place on Earth, and one that tells the time

THE QUESTION THIS CHAPTER ANSWERS

If the Earth turns once in twenty-four hours, what does that do to where a place is and to what time it is there?

IN THIS CHAPTER, YOU WILL

- √ Read and write a position as latitude and then longitude.
- √ Find a named place from its coordinates, and give the coordinates of one.
- √ Explain why the Earth's rotation makes noon happen at different moments.
- √ Work out the time in another zone from the difference in hours.
- √ Explain why crossing the International Date Line changes the date.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Royal Observatory Greenwich, Royal Museums Greenwich. The observatory the Prime Meridian runs through, and the reason the world agreed to measure longitude and time from this one place.

`www.rmg.co.uk/royal-observatory`

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

BEFORE YOU READ

Your teacher will read out a pair of numbers. Using an atlas, find the place they name. Then, without telling anybody, choose a place yourself and write down its numbers for somebody else to find. If they end up somewhere else, work out together which of you made the mistake, and what kind of mistake it was.

CHAPTER 2 · TOPIC 2.1

Latitude

The account

Latitude measures how far north or south of the Equator a place is. It is an angle, not a distance, and the angle is measured at the centre of the Earth between the Equator and the place itself.

The scale runs from zero at the Equator to ninety degrees at each pole, and it never goes further, because ninety degrees is already straight up. Every value carries a direction, north or south, and the direction is part of the reading. Forty degrees north and forty degrees south are two entirely different places with, as it happens, opposite seasons.

Lines of equal latitude are called parallels, and the name is accurate: they never meet. They do, however, get smaller as they approach the poles, because they are circles drawn round a ball rather than round a cylinder.

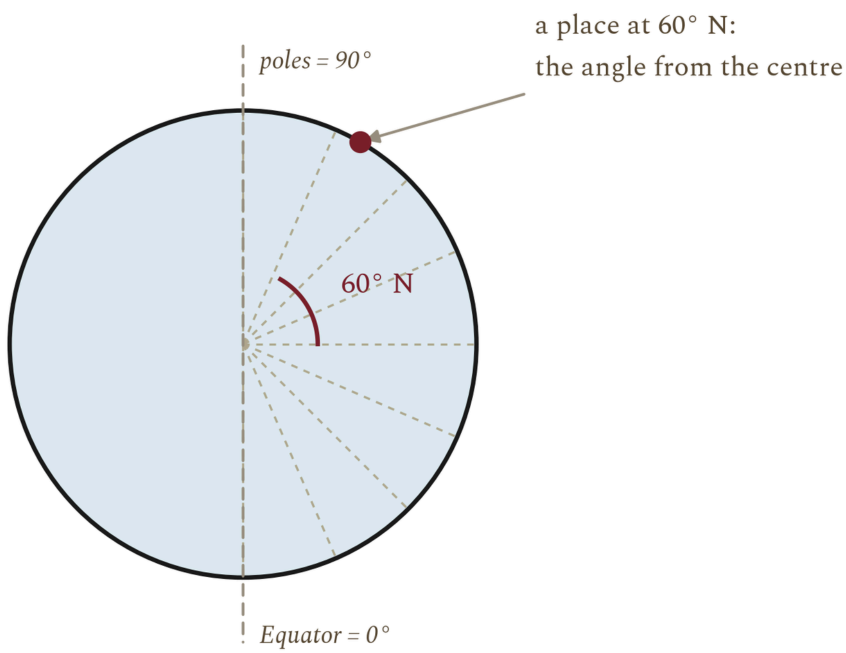


Figure 2.1 · The latitude scale, from nought to ninety.
Notice that the parallels are equally spaced in angle but not in the width of the circle they draw. The Equator is a full-sized circle; the parallel at sixty degrees is half the width.

Latitude is the single most useful number in physical geography, because it predicts so much. It tells you roughly how much solar energy a place receives across a year, which tells you roughly how warm it is, which goes a long way toward telling you what will grow there. The general rule is simple: the lower the latitude, the more direct the solar energy, and the warmer the place is likely to be.

It is a rule with exceptions, and the exceptions are interesting rather than inconvenient. Altitude changes it, because air cools as it rises: Quito sits almost on the Equator and is cool all year because it is high in the Andes. Ocean currents change it, because moving water carries heat: western Europe is far milder than its latitude alone would suggest. You will meet both again in chapter 3.

CAREFUL

Latitude is written before longitude, always, and each carries its letter. "38.7 N, 9.1 W" is Lisbon. "9.1 N, 38.7 E" is a spot in Ethiopia. The numbers are the same; the order and the letters are the whole meaning.

YOUR TURN

1. What does latitude measure, and between what two limits?
2. Why do parallels get smaller toward the poles when meridians do not?
3. Give the general rule linking latitude and temperature, then give one exception and explain it.
4. Two places are both at forty-two degrees. One is warm in July, the other cold. Explain how both can be true.

CHAPTER 2 · TOPIC 2.2

Longitude

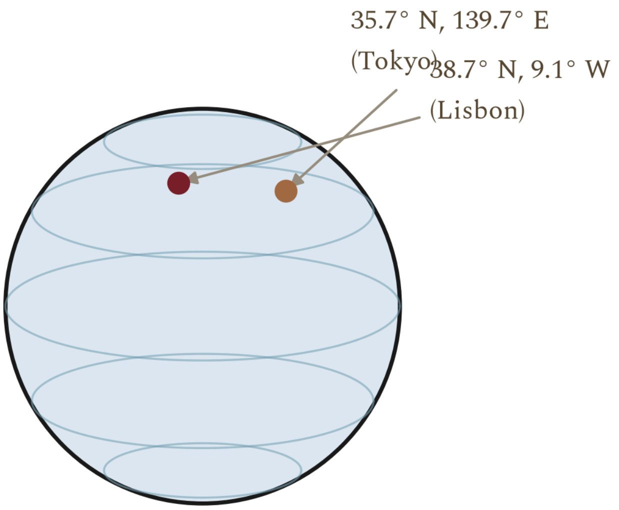
The account

Longitude measures how far east or west of the Prime Meridian a place is. Like latitude it is an angle, and like latitude it carries a direction, but its scale runs to one hundred and eighty degrees rather than ninety, because it has to cover a full circle rather than a half one.

Going east from Greenwich, the readings rise to one hundred and eighty degrees east. Going west, they rise to one hundred and eighty degrees west. Both journeys arrive at the same meridian on the opposite side of the planet, which is why that line can be written either way.

The practical difficulty with longitude, for most of history, was not understanding it but measuring it. Latitude can be found from the height of the Sun at noon or of the Pole Star at night, with an instrument and a table, and sailors could do it reasonably well by the fifteenth century. Longitude cannot be read off the sky in the same way. It has to be worked out by comparing the local time where you are with the time at a known meridian, which means carrying an accurate clock through months of damp, heat, cold and pitching decks.

That problem was severe enough that governments offered prizes for solving it. It was solved in the eighteenth century by marine chronometers accurate enough to keep reference time on a long voyage. Notice what that means: **longitude and time are the same problem.** The next five topics are all consequences of that one sentence.



Two numbers, one place: latitude fixes the parallel, longitude the meridian.

Figure 2.2 · Two numbers, one place.
Follow the parallel and the meridian for each city until they cross. The crossing is the only point on Earth with that pair of readings.

YOUR TURN

1. What does longitude measure, and what is its maximum value?
2. Why can the one hundred and eightieth meridian be described as either east or west?
3. Explain why finding longitude at sea was so much harder than finding latitude.
4. In one sentence, explain the link between longitude and time.

CHAPTER 2 · TOPIC 2.3

Absolute location

The account

Put the two measurements together and you have an absolute location: a pair of numbers that names one point on the surface of the Earth and no other. Latitude first, then longitude, each with its direction.

An absolute location does not depend on anything else being known. That is what makes it different from a relative location, which describes a place by its relationship to another place: "twenty kilometres north of Coimbra", "opposite the school", "the third house past the church". Relative locations are how people actually talk, and they are useless to somebody who does not already know the reference point.

Some absolute locations worth knowing, and worth checking in an atlas rather than taking on trust:

PLACE	LATITUDE	LONGITUDE
Lisbon	38.7 N	9.1 W
Porto	41.1 N	8.6 W
Funchal, Madeira	32.6 N	16.9 W
Ponta Delgada, Azores	37.7 N	25.7 W
London	51.5 N	0.1 W
New York	40.7 N	74.0 W
Tokyo	35.7 N	139.7 E
Sydney	33.9 S	151.2 E

Table 2.1 · Eight places, sixteen numbers.
Read down the latitude column first: four of these cities sit within four degrees of each other and are scattered across three continents. Latitude alone is never enough.

Look carefully at the Portuguese entries. Lisbon and Porto differ by less than three degrees of latitude, which is a small difference, and yet chapter 3 will show you that their climates differ noticeably. The Azores are more than sixteen degrees of longitude west of Lisbon, which is why they keep a different clock from the mainland, and, as chapter 18 will show, why they mattered enormously in a war fought across the Atlantic.

YOUR TURN

1. Define absolute location and relative location, and give an example of each for your own school.
2. Using the table, which two of the eight places are furthest apart in longitude?
3. Ponta Delgada is at 25.7 W and Lisbon at 9.1 W. How many degrees of longitude separate them?
4. Why might a rescue service insist on an absolute location rather than a description?

CHAPTER 2 · TOPIC 2.4

The Earth's rotation

The account

The Earth turns once on its axis in about twenty-four hours, and it turns from west to east. Everything in this topic follows from those two facts.

Because the Earth turns eastward, the Sun appears to move westward across the sky. It rises in the east, because that side of you is turning to face it, and sets in the west, because that side is turning away. The Sun is not going anywhere. You are.

At any instant, the Sun lights exactly half the planet. The line between the lit half and the dark half sweeps steadily westward across the surface, and as it passes a place, that place has dawn or dusk. Noon happens at a place when the Sun is highest in its sky, and that moment arrives at different instants in different places, because they are not all facing the Sun at once.

That is the whole origin of time differences. It is not a convention imposed on the world. It is a description of a spinning ball.

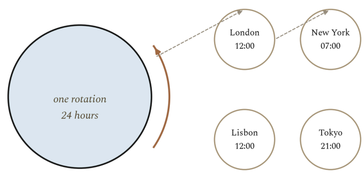


Figure 2.3 · From one rotation to four different clocks.
Read the four boxes in order. The first three are physics; only the fourth is a decision people made.

Before railways, this caused nobody any trouble. Every town simply set its clocks so that noon was noon there, and since travelling between two towns took hours or days, a difference of a few minutes was invisible. Railways destroyed that. A train timetable has to say when the train leaves, and if every station keeps its own time the timetable means nothing. Britain adopted a single railway time in the 1840s, and the rest of the industrial world followed. The next topic is what they settled on.

YOUR TURN

1. In which direction does the Earth turn, and how does that explain sunrise in the east?
2. What fraction of the Earth is lit at any instant?
3. Explain why noon does not happen everywhere at once.
4. Why did railways make local time impossible to keep?

CHAPTER 2 · TOPIC 2.5

Time zones

The account

If the Earth turns three hundred and sixty degrees in twenty-four hours, then it turns fifteen degrees in one hour. That single division is the basis of the whole system:

360 degrees ÷ 24 hours = 15 degrees per hour

So the world is divided into zones roughly fifteen degrees of longitude wide, and all the clocks inside a zone are set to the same time. Move one zone east and clocks read one hour later; one zone west, one hour earlier.

Roughly is doing real work in that sentence. Zone boundaries are drawn by governments, not by geometry, and they bend around borders, islands and convenience. Some countries that are wide enough for several zones keep only one, so that the whole country shares a working day. Some zones are half an hour or three quarters of an hour off the pattern.

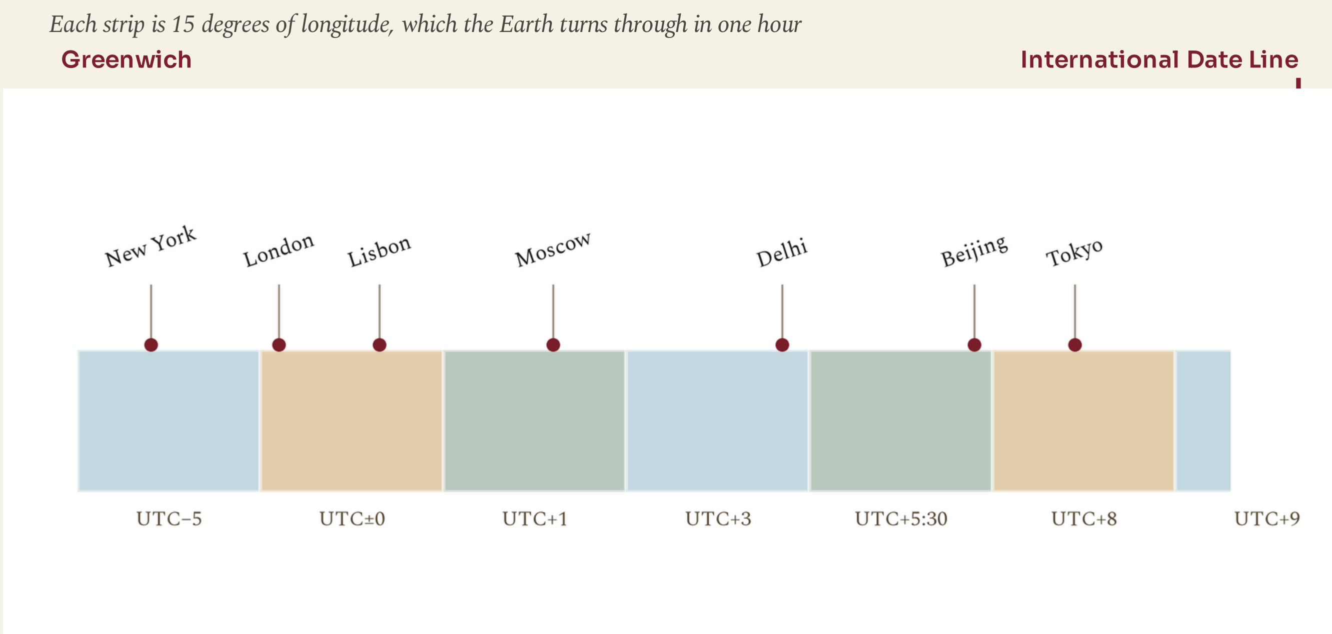


Figure 2.4 · The zones, and ten cities placed in the one their clocks really keep.
Read the offsets along the bottom. Then check a city you know against the strip it is standing in.

Here is the standard piece of arithmetic, done slowly.

WORKED ENQUIRY

It is 12:00 in London. What time is it in Tokyo, New York and Sydney?

Work from the offsets, not from a feeling about which is "ahead".

- London in winter keeps UTC, so its offset is 0.
- Tokyo keeps UTC+9. The difference is 9 minus 0, which is +9. So 12:00 plus 9

hours is **21:00 in Tokyo**, the same day.

- New York keeps UTC−5. The difference is −5 minus 0, which is −5. So 12:00

minus 5 hours is **07:00 in New York**, the same day.

- Sydney keeps UTC+10 in its winter. The difference is +10, so **22:00 in

Sydney**.

And Lisbon, which keeps UTC in winter like London, reads **12:00**: the same as London, and one hour behind Madrid, which is only a short distance east but keeps UTC+1.

CAREFUL

Many countries move their clocks forward in summer, so the offsets above change for part of the year. Always check whether summer time is in force before you answer, and say which you have assumed. A question that does not tell you should be answered for standard time, with a note.

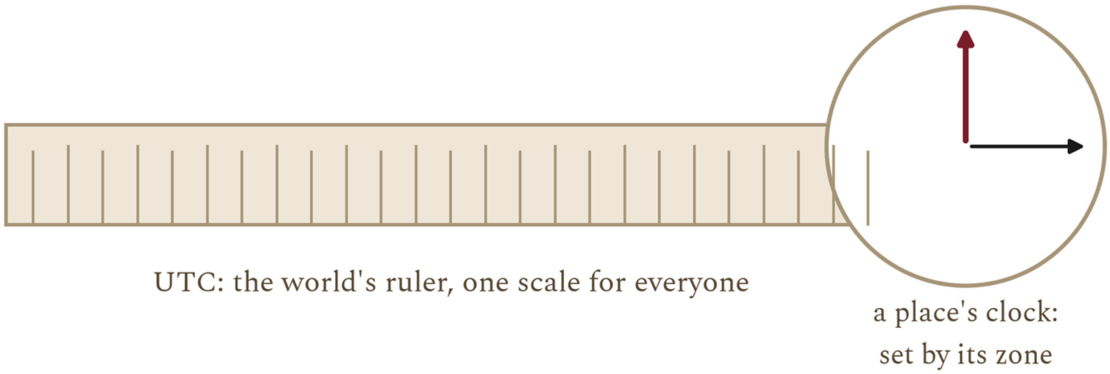


Figure 2.5 · One is a place's clock, the other is the world's ruler.
Read the last row first: it is the difference that actually matters in practice.

Greenwich still matters, and not only for history. The Prime Meridian is still at zero, longitude is still measured from it, and UTC is still kept close to the Sun's position over that meridian. When the Earth's rotation drifts very slightly out of step with atomic time, a leap second can be added to keep them together. The world runs on atomic clocks, but it still checks them against the sky.

ON THE MAP

Portugal keeps UTC in winter and UTC+1 in summer. Spain, immediately to the east, keeps UTC+1 and UTC+2. Look at where the two countries sit in longitude and explain why Portugal's choice fits the Sun better than Spain's does. Then suggest a reason a country might choose a zone that does not fit the Sun.

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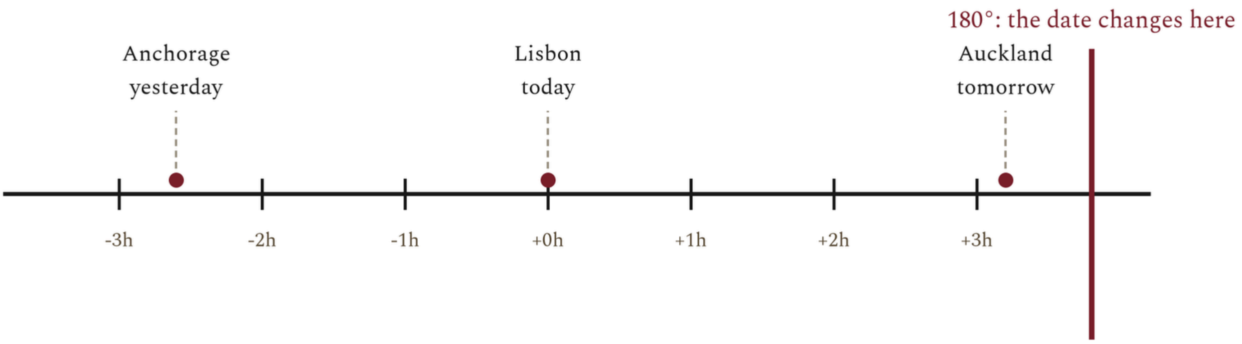


West of the line you are a day ahead; east of it, a day behind.

Figure 2.6 · Where the date changes, and four places that live with it.
Notice that the line is not the same thing as a time-zone boundary. Every zone boundary changes the hour; only this one changes the day.

FROM THE ARCHIVE

In 2011 Samoa moved from the eastern side of the line to the western side, to share working days with Australia and New Zealand rather than with the United States. To do it, the country skipped 30 December 2011 entirely. For Samoa that date does not exist: the calendar ran from Thursday 29 December straight to Saturday 31 December.

YOUR TURN

1. Explain, using the twelve-hour figures, why a date line has to exist.
2. What happens to the date when you cross the line travelling west?
3. Why does the line bend rather than run straight down the meridian?
4. Look again at the message task you did before this chapter. Rewrite "Meet me at the beach" as an instruction that would work anywhere in the world, and say what each part of it does.

WHAT YOU CAN NOW DO

- √ Read and write any position as latitude and then longitude, with directions.
- √ Explain why the Earth's rotation makes noon arrive at different moments.
- √ Calculate the time in another zone from its offset, and show the working.
- √ Explain the difference between UTC and a time zone.
- √ Explain why the International Date Line exists and why it is not straight.

WORDS FOR THIS CHAPTER

latitude, longitude, absolute location, relative location, rotation, axis, time zone, offset, Coordinated Universal Time, Greenwich Mean Time, summer time, International Date Line, meridian, parallel, chronometer

CHAPTER 2 AT A GLANCE

Latitude, longitude and time zones

if the Earth turns once in twenty-four hours, what does that do to where a place is and to what time it is there?

THE TOPICS YOU HAVE COVERED

2.1	Latitude
2.2	Longitude
2.3	Absolute location
2.4	The Earth's rotation
2.5	Time zones
2.6	Coordinated Universal Time
2.7	The International Date Line

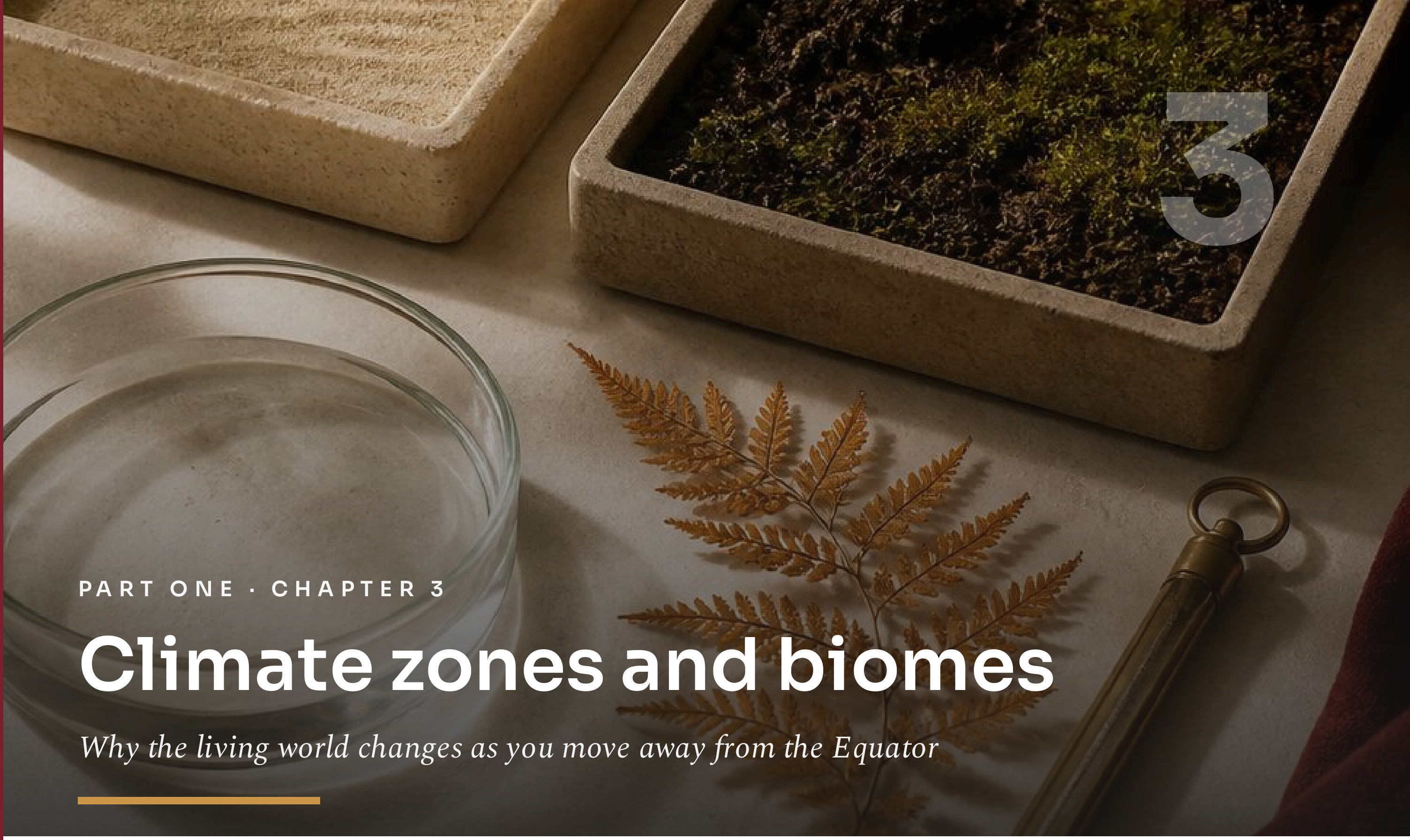
WHAT YOU CAN NOW DO

- √ Read and write any position as latitude and then longitude, with directions.
- √ Explain why the Earth's rotation makes noon arrive at different moments.
- √ Calculate the time in another zone from its offset, and show the working.
- √ Explain the difference between UTC and a time zone.
- √ Explain why the International Date Line exists and why it is not straight.

THE WORDS THIS UNIT TAUGHT

latitude · longitude · absolute location · relative location · rotation · axis · time zone · offset · Coordinated Universal Time · Greenwich Mean Time · summer time · International Date Line · meridian · parallel · chronometer

Use this page to revise before you go on. If you cannot do one of the things on the right, go back to the topic that taught it, not to the answers.



PART ONE · CHAPTER 3

Climate zones and biomes

Why the living world changes as you move away from the Equator

THE QUESTION THIS CHAPTER ANSWERS

Why is the living world so different from one latitude to another?

IN THIS CHAPTER, YOU WILL

- √ Tell weather, climate and biome apart, and use each word correctly.
- √ Explain the chain from latitude through solar energy to vegetation.
- √ Read a climate graph and name the zone it belongs to.
- √ Describe the tropical, temperate, desert and polar zones and compare them.
- √ Give examples of climate shaping farming, building and travel.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Encyclopaedia Britannica, biomes. A world biome map with each type described. Check it against the chain this chapter teaches, from latitude to vegetation.

www.britannica.com/science/biome

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

Figure 3.1 · Three words, three different jobs.
The commonest mistake in this chapter is using the first word when you mean the second. Weather changes; climate is the pattern; a biome is the living result.

Now the chain that this whole chapter is built on. It starts with position and ends with people, and every link is a cause of the next.

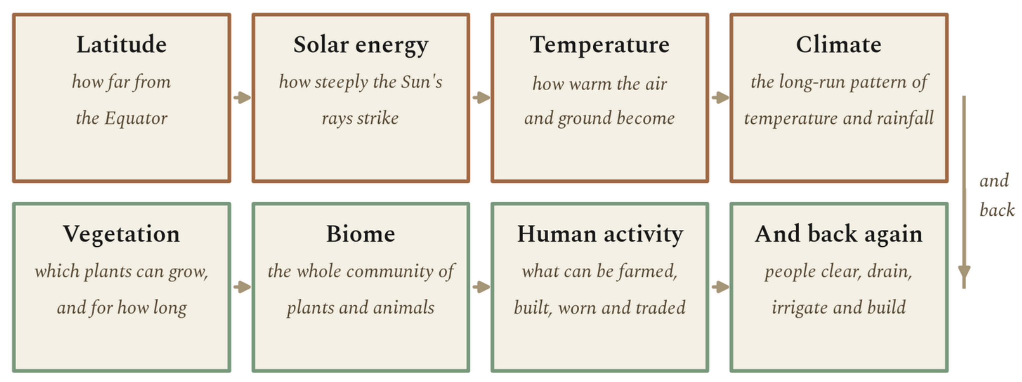


Figure 3.2 · From an angle to a way of life.
Read the boxes in order. The last arrow points back, because people change the ground as well as living on it.

The first link deserves care, because it is the one pupils most often get wrong. Latitude controls temperature not because low latitudes are nearer the Sun, but because of the **angle** at which sunlight arrives. A beam of sunlight has a fixed width. Arriving steeply, it lands on a small patch and delivers a lot of energy per square metre. Arriving at a shallow angle, the same beam is spread across a much wider patch. Shallow-angle light also travels further through the atmosphere before it lands, which absorbs and scatters more of it.

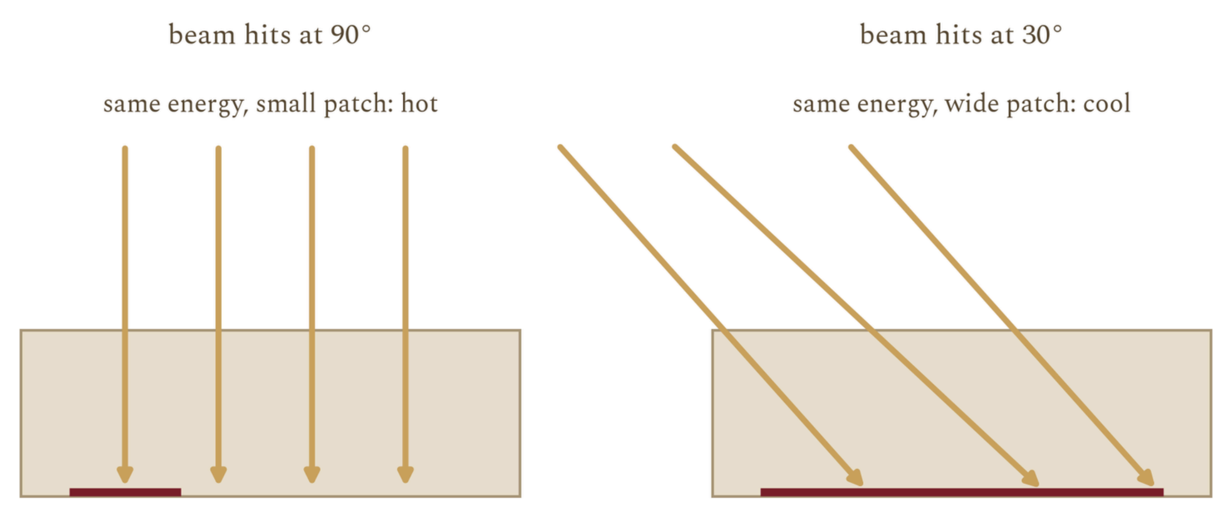


Figure 3.3 · The same beam, two different results.
Compare the width of the ground each set of rays covers. Nothing about the Sun has changed between the left and the right of this figure.

CAREFUL

Latitude is the strongest single control on climate, but it is not the only one. Altitude, distance from the sea, ocean currents, prevailing winds and mountain ranges all modify it. That is why two places at the same latitude can have very different climates, and why you should never explain a climate from latitude alone.

■ **YOUR TURN**

1. Give one-sentence definitions of weather, climate and biome.
2. Write out the chain from latitude to human activity, in order.
3. Explain, using angles, why low latitudes receive more energy per square metre.
4. Name three things other than latitude that affect a place's climate.

CHAPTER 3 · TOPIC 3.2

Tropical zones

The account

The tropical zone lies between the Tropic of Cancer and the Tropic of Capricorn. Across the whole of it, temperatures are high all year and vary little from month to month. What varies, and varies enormously, is rainfall, and rainfall is what divides the zone into different biomes.

Where rain falls heavily in every month, you get **tropical rainforest**. The Amazon basin, the Congo basin and the islands of south-east Asia are the three great blocks of it. The forest grows in layers, with a closed canopy overhead and very little light reaching the ground. It is the most biodiverse land environment on Earth: a single hectare can hold more tree species than the whole of temperate Europe.

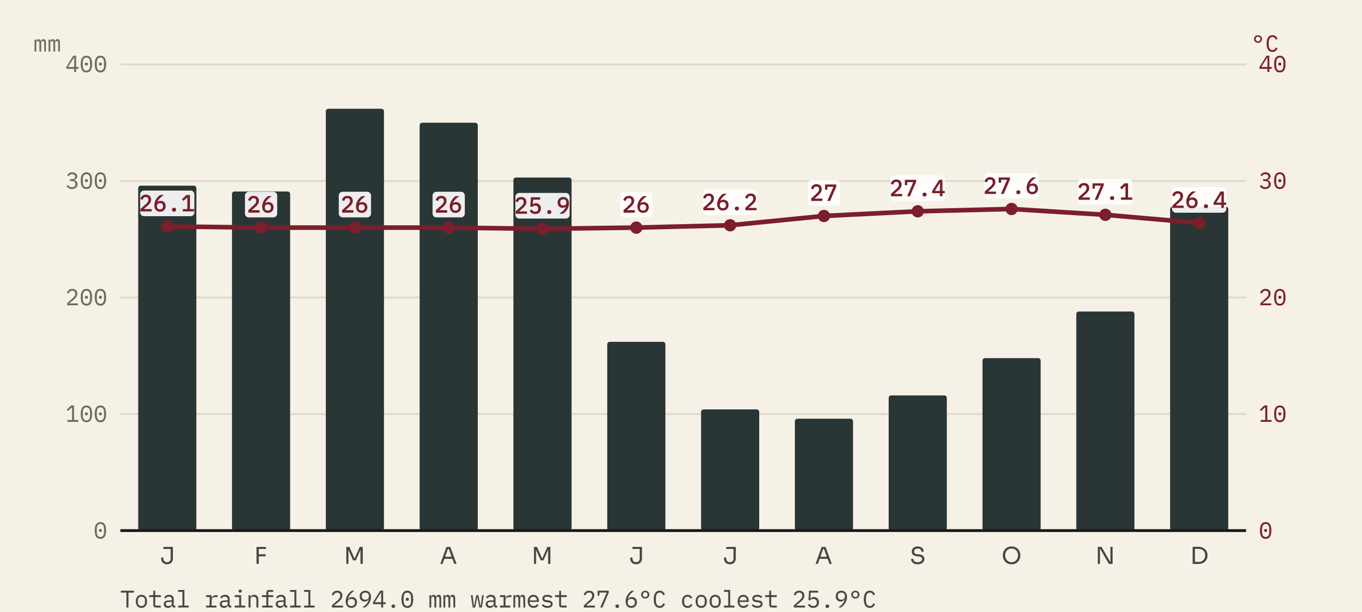


Figure 3.4 · Manaus, in the Amazon basin, 1991 to 2020.
Read the bars first: every month is wet, and even the driest is wetter than Lisbon's wettest. Then read the line: the warmest and coolest months differ by less than two degrees. This is what "warm all year" looks like on a graph.

Where the rain comes in a marked wet season followed by a marked dry season, you get **savanna**: grassland with scattered fire-resistant trees, across much of Africa, northern Australia and central Brazil. The grasses grow fast in the wet season and die back in the dry, and fire is a normal part of the system rather than a disaster in it.

There is a counter-intuitive point here that is worth stopping on. Rainforest soils are usually **poor**, not rich. Almost all the nutrients are locked up in the living plants, and the constant heavy rain washes minerals out of the soil. When forest is cleared for farming, the ash from burning gives one or two good harvests and then the soil is exhausted. That is a geographical fact with very large historical consequences, and you will meet it again in chapter 12.

ON THE MAP

On a world map, shade the rainforest belt along the Equator and the savanna belts on either side of it. Then mark the Amazon, Congo and south-east Asian blocks. Which of the three is the largest, and which countries share it?

YOUR TURN

1. What is constant across the tropical zone, and what varies?
2. What is the difference between rainforest and savanna, and what causes it?
3. Using the Manaus graph, give the wettest month, the driest month and the annual total.
4. Explain why rainforest soils are poor, and why that matters to farming.

CHAPTER 3 · TOPIC 3.3

Temperate zones

The account

The temperate zones lie between the Tropics and the polar circles, in both hemispheres. Their defining feature is not mildness but **seasons**: a real summer and a real winter, with noticeable differences in temperature and in day length between them.

Within the temperate zone there are several distinct climates, and three matter for this book.

Temperate forest climates have rain spread fairly evenly through the year and cold but not extreme winters. Trees are often deciduous, dropping their leaves in autumn because keeping them through a cold, dark winter costs more than it earns. Much of western and central Europe belongs here, and almost all of it was forest before people cleared it for farming.

Temperate grassland sits in continental interiors, far from the sea, where rainfall is lower. The Eurasian steppe, the North American prairie and the Argentine pampas are the great examples. These are the world's grain lands, and their deep, dark soils are among the most productive on the planet.

Mediterranean climates are the odd one out, and they are the ones a Portuguese pupil lives in. They have mild wet winters and hot dry summers, which is the reverse of the usual pattern of rain falling when plants most want it. Plants there are adapted to drought in the growing season: small tough leaves, deep roots, thick bark. Olive, cork oak, holm oak, vine and citrus are all Mediterranean plants, and all five are grown in Portugal.

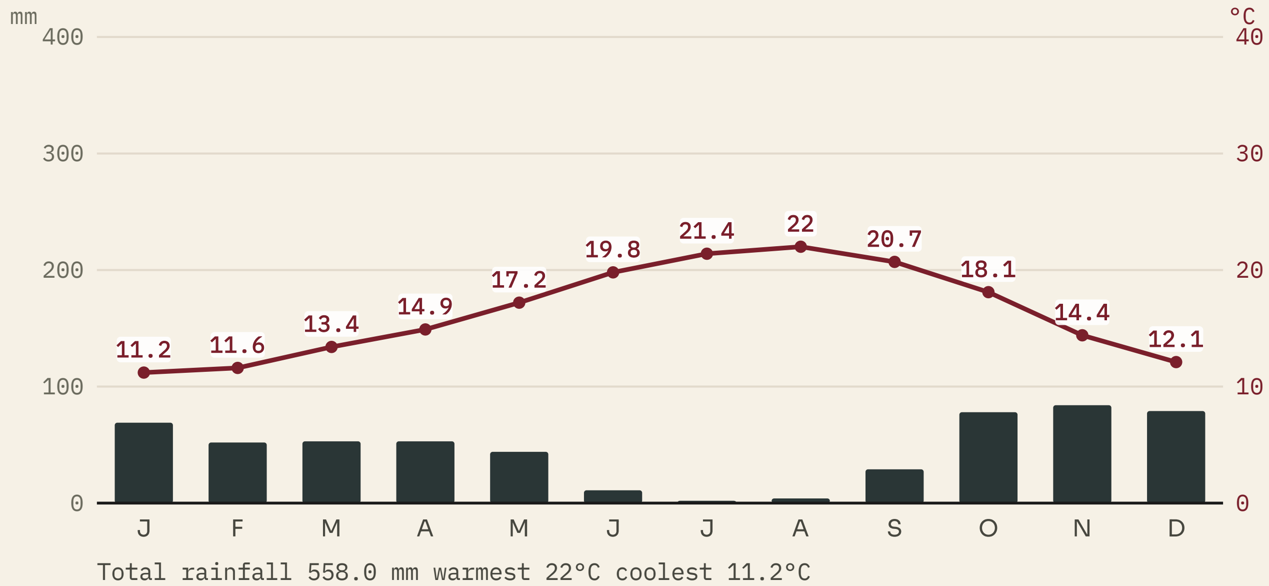


Figure 3.5 · Lisbon, 1991 to 2020.
Look at July and August: almost no rain at all, in the hottest months of the year. Then look at November and December. A Mediterranean climate delivers its water when the plants least need it, which is why the vegetation there is built to wait.

WHERE PORTUGAL FITS

Portugal is a good place to see that latitude is not the whole story. Porto is only about two and a half degrees north of Lisbon, yet it is noticeably wetter, because it faces the Atlantic more directly and catches more of the westerly weather systems. Move inland to Braganca and the summers are hotter and the winters colder, because the sea is too far away to moderate either. The same country, a few degrees, three noticeably different climates. Latitude, distance from the ocean and altitude are all at work at once.

YOUR TURN

1. What single feature defines the temperate zones?
2. Name the three temperate climates in this topic and give one example region for each.
3. Using the Lisbon graph, give the driest month and the annual rainfall total.
4. Explain why Mediterranean plants have small, tough leaves.

CHAPTER 3 · TOPIC 3.4

Desert regions

The account

A desert is defined by **rainfall**, not by temperature. The usual threshold is under about 250 millimetres a year. That is the whole definition. It says nothing at all about heat, which is why the largest desert on Earth is Antarctica.

Most hot deserts sit in two belts, at roughly twenty to thirty degrees north and south. That is not a coincidence either. Air that rose at the Equator, dropped its rain and travelled poleward high in the atmosphere sinks back down around those latitudes, and sinking air warms and dries. So the desert belts are the other end of the same circulation that makes the rainforest belt.

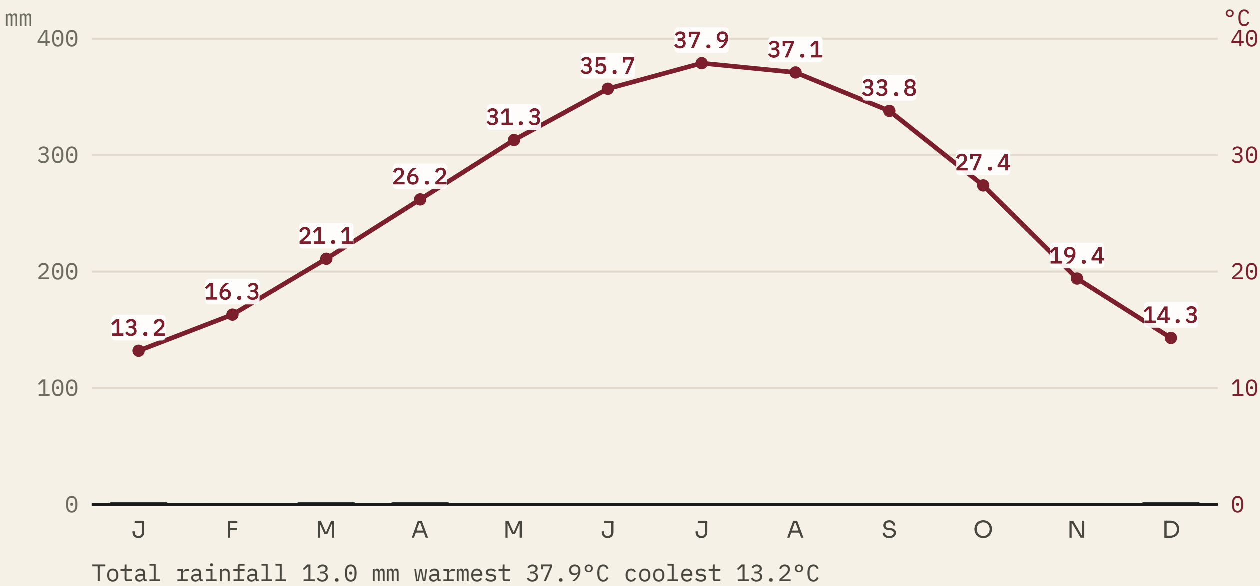


Figure 3.6 · In Salah, in the Algerian Sahara, 1991 to 2020.
The rainfall bars are almost invisible, and that is the point: the annual total is a fortnight of Lisbon's. Notice also the temperature range across the year, which is far wider than at Manaus.

Cold deserts work the same way for a different reason. The Gobi and the dry interior of central Asia are far from any ocean, so little moisture reaches them; some are also in the rain shadow of mountains, which strip the moisture out of air before it arrives. They are as dry as hot deserts and bitterly cold in winter.

	Tropical rainforest	Hot desert
Rainfall	More than 2000 mm a year, in every month.	Under 250 mm a year, and some years none at a
Temperature	Warm and almost unchanging: every month above 18 °C.	Very hot by day, and often cold at night.
Vegetation	Layered forest, evergreen, growing all year.	Sparse, spaced out, built to store water and lose little.
Soil	Thin and poor. The nutrients are in the living plants.	Thin, stony, and often salty where water evapor
People	Farming clears it, and the cleared soil is soon exhausted.	Settlement follows water: oases, wells and river from elsewhere.

Figure 3.7 · Two dry places, and what makes each of them dry.
Read the rainfall row first, because that is the row that makes both of them deserts. Everything else about them differs.

Life in a desert is built round water. Plants space themselves widely so that each has enough ground to draw from, store water in stems or leaves, and often lose their leaves entirely in the driest months. People settle where water is: at oases, over aquifers, or along rivers that rise somewhere wetter and simply pass through. Egypt is the classic case, a narrow ribbon of dense settlement along a river running through one of the driest places on Earth.

YOUR TURN

1. Give the definition of a desert. Which word in your answer is doing the work?
2. Explain why hot deserts cluster near twenty to thirty degrees of latitude.
3. Give two reasons a cold desert can be dry.
4. Using the In Salah graph, give the annual rainfall total, and compare it with Lisbon's.

CHAPTER 3 · TOPIC 3.5

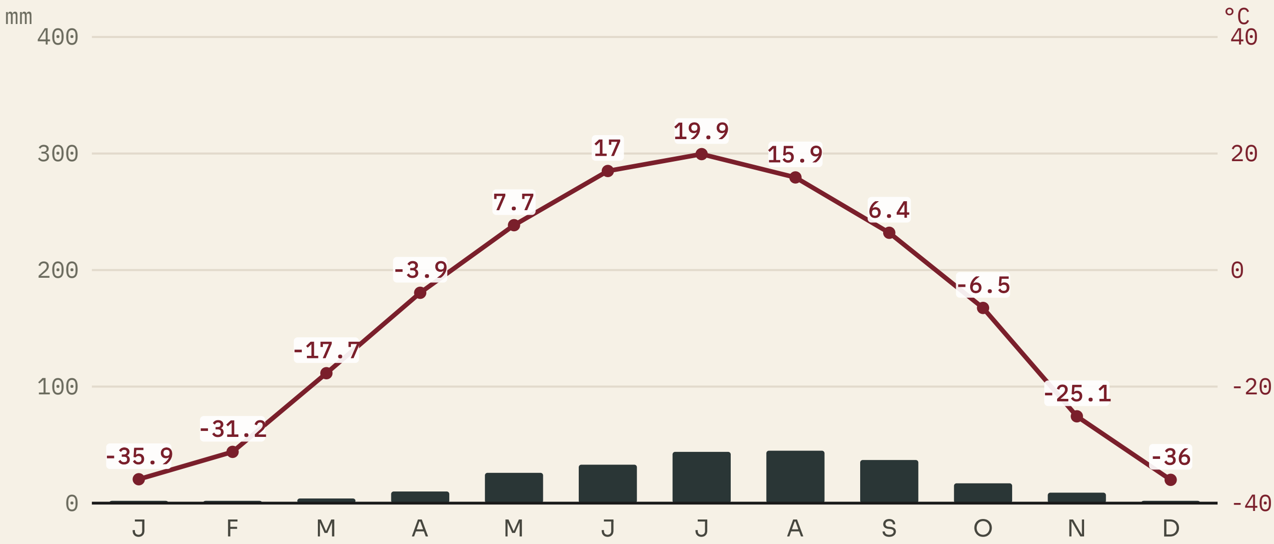
Polar zones

The account

Beyond the polar circles, the defining conditions are cold and an extreme swing in daylight. Growing seasons are very short: in some places the ground is suitable for plant growth for only six to ten weeks a year.

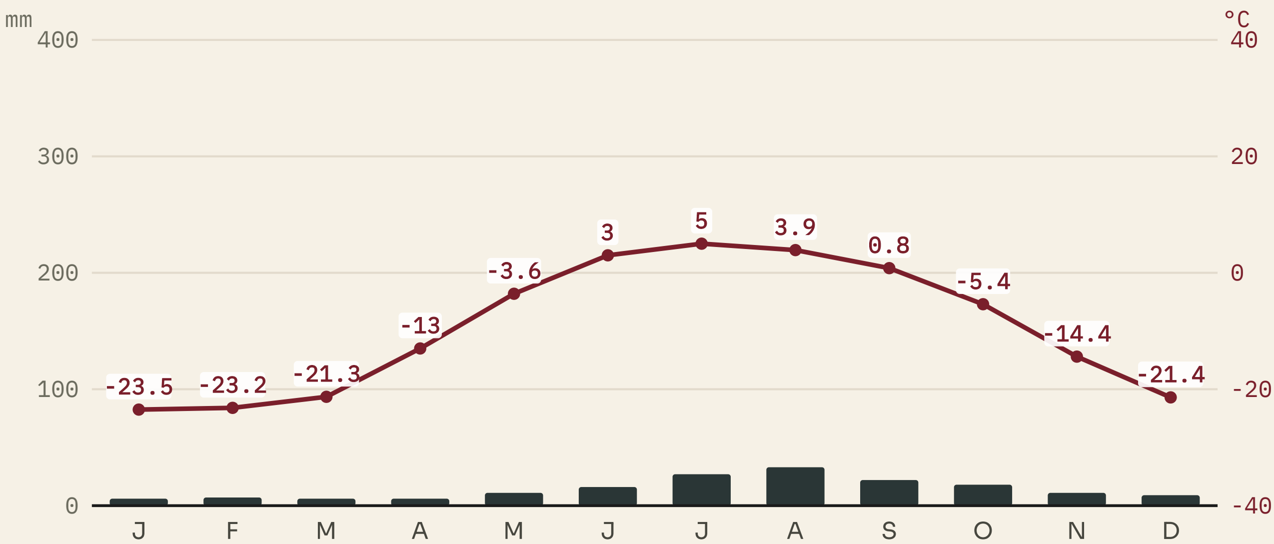
Tundra is the biome of the ground that thaws at the surface in summer. Beneath it lies **permafrost**, ground that has stayed below freezing for at least two years and often for far longer. Because water cannot drain through frozen ground, tundra in summer is boggy, and because the growing season is so short, plants stay low: mosses, lichens, sedges, dwarf shrubs and, further south, stunted trees. There are no tall trees, because a tall tree cannot get water through frozen ground and cannot survive the wind.

South of the tundra, in the northern hemisphere, is the **taiga** or boreal forest, the largest forest on Earth, running across Canada, Scandinavia and Siberia. It is coniferous almost throughout: needles rather than leaves, because needles lose less water and can start work as soon as the light returns.



Total rainfall 231.0 mm warmest 19.9°C coolest -36°C

Figure 3.8 · Yakutsk, in eastern Siberia, 1991 to 2020.
This is the largest annual temperature range in this chapter: look at the difference between January and July. The rainfall total is small enough that some would call it a cold desert, which shows how the categories overlap at their edges.



Total rainfall 172.0 mm warmest 5°C coolest -23.5°C

Figure 3.9 · Utqiagvik, on the Arctic coast of Alaska, 1991 to 2020.
Compare it with Yakutsk. It is far less extreme in winter, because it sits beside an ocean, and far cooler in summer, for the same reason. Distance from the sea is doing as much work here as latitude.

Adaptations are severe and specific. Animals grow thick insulating coats, change colour with the season, store fat, hibernate or migrate. Human settlement has historically depended on hunting, fishing and herding rather than on crops, because crops need a growing season the ground does not provide.

YOUR TURN

1. What is permafrost, and why does it make tundra boggy in summer?
2. Why are there no tall trees on the tundra? Give two reasons.
3. Compare the Yakutsk and Utqiagvik graphs. Which has the greater annual range, and why?
4. Explain why polar peoples have historically herded, hunted and fished rather than farmed.

CHAPTER 3 · TOPIC 3.6

Climate and human activity

The account

Climate does not decide what people do. It changes what things cost, and cost decides a great deal.

Farming follows climate most directly, because plants do. Rice needs warmth and standing water, so it is grown across monsoon Asia. Wheat needs a cooler, drier ripening season, so it belongs to temperate grasslands. Olives and vines need dry summers, so they belong to Mediterranean lands, which is why Portugal grows them and Norway does not.

Building follows climate too, and you can read the climate off a roof. Steep roofs shed snow, so they belong where snow falls. Flat roofs belong where it never does, and are used as working space. Thick walls and small windows keep heat out in hot dry lands and heat in in cold ones. Traditional Alentejo houses are painted white to reflect the summer sun and have small windows for the same reason.

Travel and trade follow climate in ways that turn up throughout the rest of this book. Rivers that freeze in winter close a trade route for months. Mountain passes fill with snow. Seas ice over. The monsoon winds of the Indian Ocean reverse twice a year, and for centuries the whole timetable of trade between east Africa, Arabia and India was set by them: sail one way in one season, wait, and sail back in the other.

What can be grown
Rice needs warmth and standing water, wheat needs a cool, dry ripening season.

→

Where people build
Roofs are steep where snow falls and flat where it never does.

→

How places connect
Rivers that freeze, passes that close and seas that ice over all set the timetable.

Figure 3.10 · Three ways a climate shows up in a life.
None of these is a rule about what people must do. Each is a statement about what is cheap and what is expensive where they live.

GO FURTHER

Climate sets costs, and people change those costs. Irrigation lets crops grow where rainfall would not allow them. Greenhouses move a climate indoors. Cheap refrigerated shipping means a country can eat food that could never be grown in it. So the honest statement is not "climate decides", nor "climate does not matter", but "climate sets the price, and technology and money can pay it". Keep that formula. You will need it in chapter 6, when you are asked why industry began in one country rather than another.

YOUR TURN

1. Give one crop each for a tropical, a temperate and a Mediterranean climate, and say why each belongs there.
2. What can you tell about a place's climate from the shape of its roofs?
3. Explain how the monsoon set the timetable for Indian Ocean trade.
4. "Climate decides how people live." Rewrite that sentence so that it is accurate, and explain what you changed.

WHAT YOU CAN NOW DO

- √ Use the words weather, climate and biome correctly and separately.
- √ Explain the chain from latitude through solar energy to human activity.
- √ Read a climate graph and say which zone it belongs to and why.
- √ Compare the tropical, temperate, desert and polar zones on real data.
- √ Explain how climate sets the cost of farming, building and travel.

WORDS FOR THIS CHAPTER

weather, climate, biome, solar energy, rainforest, savanna, temperate forest, grassland, Mediterranean climate, desert, rain shadow, tundra, permafrost, taiga, growing season, biodiversity

CHAPTER 3 AT A GLANCE

Climate zones and biomes

why is the living world so different from one latitude to another?

THE TOPICS YOU HAVE COVERED

3.1	Latitude and climate
3.2	Tropical zones
3.3	Temperate zones
3.4	Desert regions
3.5	Polar zones
3.6	Climate and human activity

WHAT YOU CAN NOW DO

- √ Use the words weather, climate and biome correctly and separately.
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Use this page to revise before you go on. If you cannot do one of the things on the right, go back to the topic that taught it, not to the answers.

PART ONE · CHAPTER 4

Hemispheres and global patterns

Two ways to cut the world in half, and what each cut shows

THE QUESTION THIS CHAPTER ANSWERS

How alike are the two halves of the planet, whichever way you cut it?

IN THIS CHAPTER, YOU WILL

- √ Compare the northern and southern hemispheres for land, ocean and people.
- √ Explain why the seasons are reversed between the two.
- √ Say where the eastern and western hemispheres are divided, and by what.
- √ Explain why the western hemisphere and the West are not the same idea.
- √ Give the main reasons people are distributed as unevenly as they are.

THE TOPICS, IN THE ORDER YOU MEET THEM

WATCH, READ AND LISTEN

Our World in Data, world population growth. Where the world's people actually are, measured. Test the claim that the northern hemisphere holds the great majority of them.

ourworldindata.org/world-population-growth

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

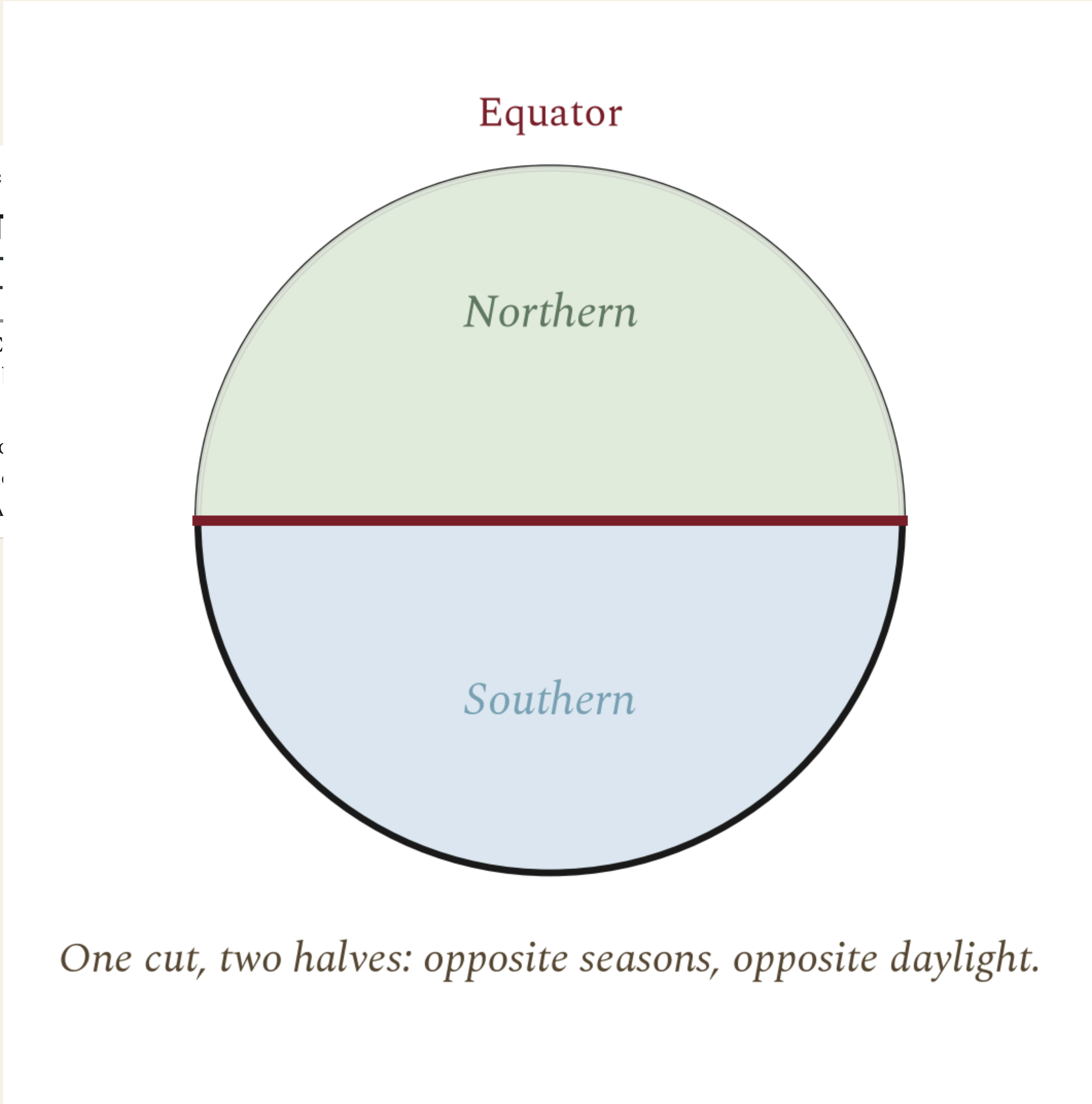


Figure 4.1 · One cut, two very different halves.
Compare the amount of land above the line with the amount below it. This single imbalance explains most of the differences in this topic.

People follow the land. Since people live on land rather than on water, and most of the land is northern, the great majority of the world's population is in the northern hemisphere. So are most of the largest cities, most of the industry and most of the world's political and economic weight, for reasons that are historical as well as geographical.

Climates behave differently. A hemisphere that is mostly ocean has milder extremes, because water heats and cools far more slowly than land does. Southern hemisphere temperate climates are generally less extreme than northern ones at the same latitude, and there is no southern equivalent of the bitter continental winter of Siberia or central Canada, because there is no southern landmass at that latitude to have one.

CAREFUL

"Most people live in the northern hemisphere" is a statement about where the land is, not a statement about wealth, development or importance. Chapter 9 and chapter 33 both deal with unevenness between countries, and neither is explained by which side of the Equator a country sits on.

YOUR TURN

1. Which hemisphere holds most of the world's land, and roughly what does the other one hold instead?
2. Explain why most people live in the northern hemisphere.
3. Why are southern temperate climates generally less extreme than northern ones at the same latitude?
4. Name one continent that lies wholly in the southern hemisphere and one that lies wholly in the northern.

CHAPTER 4 · TOPIC 4.2

The eastern and western hemispheres

The account

The second cut is made the other way, along a meridian, and this one is a human decision rather than a fact of the planet. The dividing lines are the Prime Meridian at zero degrees and the one hundred and eightieth meridian on the far side. Everything between them going east is the eastern hemisphere; everything going west is the western hemisphere.

The eastern hemisphere contains almost all of Europe, Africa and Asia, and Australia. The western hemisphere contains North and South America, and, because the Prime Meridian passes through western Europe and west Africa, a slice of those too.

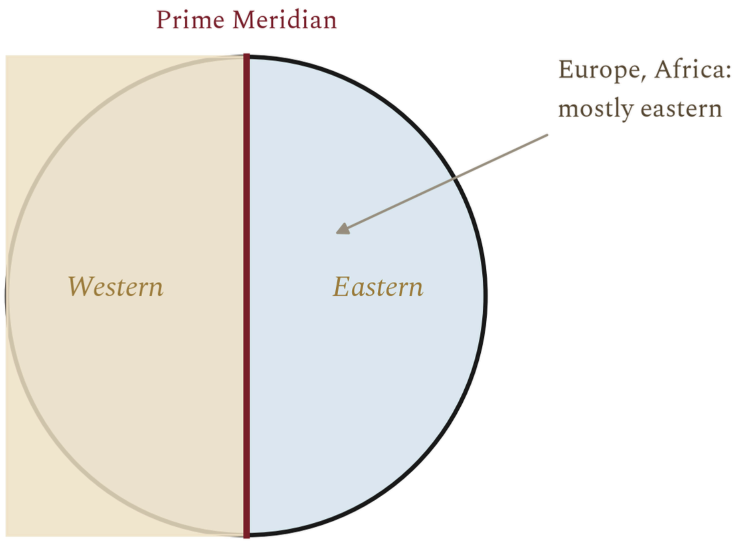


Figure 4.2 · The second cut, and where it falls.
Trace the Prime Meridian down through Britain, France, Spain, Algeria, Mali and Ghana. Everything to its left, as far as the far side of the Pacific, is the western hemisphere.

That last detail catches people out, and it is worth being exact about. Mainland Portugal lies entirely west of the Prime Meridian, so **Portugal is in the western hemisphere** in the geographical sense. So are Ireland, most of Britain, western France, Spain, Morocco, Senegal and Ghana. The line is at Greenwich, not at the Atlantic coast.

YOUR TURN

1. Which two meridians divide the eastern and western hemispheres?
2. Name three continents that lie mainly in the eastern hemisphere.
3. Is Portugal in the eastern or the western hemisphere? Explain how you know.
4. Why is this division a human decision when the Equator is not?

CHAPTER 4 · TOPIC 4.3

The geographical west and the political West

The account

Two phrases sound similar and mean completely different things, and confusing them causes a surprising amount of muddle.

The western hemisphere is a geographical term. It means everything between the Prime Meridian and the one hundred and eightieth meridian, going west. It is decided by an agreement of 1884 about where to measure longitude, and by nothing else.

The West is a political and cultural term. It is used to describe a group of countries thought of as sharing a broad political tradition, and it has changed meaning more than once. In the Cold War, which you will study in chapter 21, it meant the states aligned with the United States, whether or not they were democracies. Today it is used more loosely still.

	The western hemisphere	The West
What kind of term	Geographical. Everything west of the Prime Meridian as far as 180 degrees.	Political and cultural, and it has changed mean more than once.
Who decides it	An agreement of 1884 about where to measure longitude from.	Usage. No treaty defines it and no map fixes it.
Awkward case	Most of Africa and all of western Europe sit in it, including Portugal.	Australia and New Zealand are usually counte and they are about as far east as it is possible to be.
Why it matters	It tells you where a place is.	It tells you which side of an argument somebod thinks a country is on.

Figure 4.3 · Two phrases, two different jobs.
Read the last row first. One tells you where a place is; the other tells you which side of an argument somebody thinks it is on.

The awkward cases show the difference best. **Australia and New Zealand** are routinely counted as part of the West and are about as far east as it is possible to get. **Portugal** is in the western hemisphere geographically, and was also counted as part of the West during the Cold War, but at that time it was a dictatorship rather than a democracy, which is exactly the kind of complication chapter 21 exists to explain.

So the two terms can agree, disagree, or have nothing to do with each other. The rule for your own writing is simple: if you mean position, say hemisphere. If you mean politics, say what you actually mean and name the countries.

YOUR TURN

1. Define the western hemisphere and the West, and say what kind of term each is.
2. Give one country that is in one but not obviously in the other, and explain.
3. Why can no map settle an argument about whether a country is part of the West?
4. Rewrite this sentence so that it is precise: "Portugal has always been a western country."

CHAPTER 4 · TOPIC 4.4

Global population patterns

The account

People are not spread evenly over the Earth, and they never have been. Very large areas hold almost nobody, and small areas hold enormous numbers. Two regions alone, south Asia and east Asia, hold a large share of all the people alive.

The places that are nearly empty have things in common: they are too dry, too cold, too high, too steep, too wet underfoot, or too far from anywhere. The Sahara, the Arabian interior, central Australia, the Amazon basin, the Siberian north, Greenland, the high Himalaya and Antarctica between them cover a very large part of the land surface and hold a very small part of the population.

The places that are crowded also have things in common, and they are the reverse of that list.

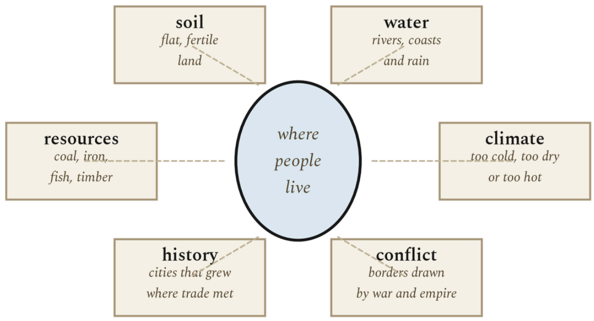


Figure 4.4 · Six reasons people are where they are.
No single one of these explains a dense population; every dense population has several of them at once. The last box is the one people forget.

The last of those six is the one this book cares about most. **People are also where earlier people were.** A city does not move because the reason it was founded has stopped applying. Ports founded for sailing ships still hold cities in the age of container terminals. Towns founded at a river crossing still stand there long after bridges made the crossing unremarkable. Population maps change slowly, which is why a geography lesson keeps turning into a history lesson.

WHERE PORTUGAL FITS

Portugal's own population map shows the pattern clearly. The coastal strip from Braga through Porto to Lisbon and Setubal is densely settled; the interior, particularly the north-east and parts of the Alentejo, is thinly settled and in places has been losing people for decades. The reasons are the six above, all at once: the coast has the water, the milder climate, the flatter usable land, the ports, the work and the historical head start. The interior has fewer of each. You will meet the consequences again in chapter 7, when industry concentrates on the same coastal strip.

GO FURTHER

Look again at the sketch you drew before this chapter. Almost everybody draws the world's people too evenly, and almost everybody underestimates Asia. Ask yourself why. One likely reason is that the maps you see most often are political maps, where every country gets the same visual weight whatever its population, and news coverage that is not distributed by population either. Being able to notice that your mental map is skewed, and in which direction, is a genuine geographical skill.

YOUR TURN

1. Name four kinds of place that hold very few people, and say why in each case.
2. Give the six factors from the figure, and rank them for your own home region.
3. Explain, with an example, why population maps change slowly.
4. Compare your first sketch with an atlas. Which part of the world did you get most wrong, and can you say why?

WHAT YOU CAN NOW DO

√ Compare the northern and southern hemispheres for land, ocean, people and

climate, and explain the differences.

√ Say exactly where the eastern and western hemispheres are divided.

√ Explain why the western hemisphere and the West are different ideas.

√ Give the main reasons the world's population is distributed as it is.

√ Explain why a population map is partly a record of history.

WORDS FOR THIS CHAPTER

hemisphere, Equator, Prime Meridian, distribution, density, population, continental climate, maritime climate, the West, sparsely populated, densely populated

PART TWO

The Renaissance and the road to industry

Four movements, one line. The Renaissance taught Europe to observe and measure. Discovery turned that measurement onto the ocean and stitched the world together. The Reformation broke the one Church into competing ideas, and print spread them. Revolutions, in America and France, handed sovereignty to citizens and an emperor. And the whole inheritance, capital, coal, machines and ideas, met in one small island and produced the Industrial Revolution, which is where the modern world begins.

5

PART TWO - CHAPTER 5

The Renaissance

Rebirth, humanism and the measure of all things

THE QUESTION THIS CHAPTER ANSWERS

What happened in Europe when it started to look at the world, and at itself, with new eyes?

THEM

IN THIS CHAPTER, YOU WILL

- √ Define the Renaissance and explain why it began in Italy.
- √ Explain perspective, anatomy and patronage in Renaissance art.
- √ Describe how printing changed the spread of ideas.
- √ Describe Portugal's own version of the Renaissance.

WATCH, READ AND LISTEN

Khan Academy, the Renaissance in Europe. A short course on the art, ideas and patrons of the period.

www.khanacademy.org/humanities/renaissance-reformation

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

5.1 What was the Renaissance?

Between the fourteenth and the sixteenth centuries, Europe changed the way it thought, painted, built and understood the world. Historians call that change the **Renaissance**, a French word meaning rebirth. What was reborn was the culture of ancient Greece and Rome, which had been studied only in fragments for a thousand years.

The story begins in Italy. After the Black Death of the 1340s killed perhaps a third of Europe's people, survivors in the wealthy cities of northern Italy, above all **Florence**, turned to the remains of the classical past. Italian scholars could read Roman inscriptions in their own streets, and trade with the eastern Mediterranean brought Greek manuscripts, and Greek scholars, westward after the fall of Constantinople in 1453.

Two ideas ran through the whole movement. The first was **humanism**: the belief that human beings, their dignity, their reason and their achievements, were worthy of study in themselves, not only as preparation for the afterlife. The second was a new confidence in individual ability, expressed most clearly in the term *uomo universale*, the universal man, who could paint, build, engineer and think.

Why Italy?

Geography, as always in this book, mattered. Italy's city-states sat on the trade routes between northern Europe and the eastern Mediterranean, so merchant families such as the **Medici** of Florence grew rich enough to become patrons, paying artists and scholars to produce work of the highest quality. Competition between cities and between patrons pushed quality higher still.

The printing press, which you will meet again in chapter 7, spread the new ideas quickly. A book copied by hand reached hundreds of readers; a printed book reached hundreds of thousands.

YOURTURN

1. Explain the meaning of the word Renaissance.
2. Give two reasons why the movement began in Italy rather than in northern Europe.
3. What did humanists believe that made them different from earlier medieval scholars?

5.2 Art and the human form

Renaissance art announced itself with a change you can see. Medieval art was symbolic: figures were flat, sized by importance, and set against gold backgrounds that stood for heaven. Renaissance art was observational: figures had weight and anatomy, and they stood in real space.

The technical key was **linear perspective**, worked out in Florence by the architect Filippo Brunelleschi around 1420 and codified by Leon Battista Alberti in 1435. By making parallel lines converge on a single *vanishing point*, a painter could represent three-dimensional depth on a flat panel with mathematical precision. Masaccio used it in his frescoes within a decade; a century later it was simply how painting worked.

The second key was anatomy. Artists such as **Leonardo da Vinci** and Michelangelo dissected bodies to understand muscle and bone. Leonardo's drawings of the human skeleton, the heart and a foetus in the womb are simultaneously works of art and works of science, and that double nature is the clearest possible expression of the Renaissance mind.

The third was patronage. The Medici in Florence, the papacy in Rome and the doges in Venice paid for altarpieces, chapels and portraits. Michelangelo's frescoes on the ceiling of the Sistine Chapel in Rome, painted between 1508 and 1512, were commissioned by Pope Julius II, himself one of the great patrons of the age.

Artist	Known for	Where to see it
Donatello	First free-standing bronze **David** since antiquity	Florence
Botticelli	*The Birth of Venus*, classical myth on canvas	Florence
Leonardo da Vinci	*Mona Lisa*, notebooks, anatomy studies	Paris
Michelangelo	Sistine Chapel ceiling, *David*	Rome, Florence
Raphael	Madonnas, frescoes in the Vatican	Rome

Table 5.1 Five artists who define the High Renaissance.

YOURTURN

1 Explain how linear perspective works, using the term vanishing point.

2 Why did artists study anatomy?

3 Why did patronage matter to the quality of Renaissance art?

5.3 Science and the printed word

The Renaissance did not only change pictures. It changed what counted as evidence. The astronomer **Nicolaus Copernicus**, working in Poland, argued in 1543 that the Earth travels round the Sun, displacing the ancient Earth-centred model of Ptolemy. He had no telescope; his evidence was patient observation and mathematics. A century later Galileo Galilei's telescope confirmed the idea, and the justified conflict with the authorities of his day showed how far the new confidence in observation would push.

Anatomy, engineering and cartography advanced on the same principle: look, measure, and trust the measurement over the authority of old books. Brunelleschi's dome for Florence cathedral, finished in 1436 and still the largest masonry dome ever built, was designed with the same mathematics that painters used for perspective.

The spread of these ideas owed everything to the press. After Johannes Gutenberg printed his Bible at Mainz around 1455, presses multiplied across Europe. By 1500, perhaps twenty million books had been printed. A discovery made in one city could be read in every other city within months, and the pace of change in art, science and politics accelerated accordingly.

YOURTURN

1. What evidence did Copernicus use, and what did he lack?
2. Explain why the printing press accelerated the Renaissance.
3. What principle did anatomy, engineering and cartography share?

5.4 Portugal in the global story: the Renaissance on the Atlantic edge

Portugal sat at the edge of the Renaissance world, and it made that position an advantage. While Florentine merchants financed altarpieces, the Portuguese crown financed navigation, and the two were the same spirit of measurement applied to different frontiers.

Prince Henry the Navigator gathered astronomers, cartographers and shipbuilders at Sagres in the Algarve and systematised the exploration of the African coast. The Escola de Sagres, as tradition names it, produced the improved caravel and the methodical, voyage-by-voyage extension of the known map. Portuguese Renaissance architecture had its own style, the Manueline, named for King Manuel I, which carved ropes, coral and armillary spheres into stone, the architecture of a nation thinking about the ocean.

In 1572 the poet Luís de Camões published *Os Lusíadas*, an epic of Vasco da Gama's voyage to India. It is the one Renaissance epic whose hero is a navigator rather than a warrior, and it is studied wherever the period is taught.

YOURTURN

1. How did the Portuguese Renaissance differ in emphasis from the Italian?
2. What is Manueline architecture, and what do its carvings represent?
3. Why is Os Lusíadas unusual among Renaissance epics?

WORDSFORTHISCHAPTER

Renaissance, the rebirth of classical learning and art from the fourteenth to the sixteenth century; **humanism**, the study of human achievement, classical texts and human dignity; **patron**, a wealthy person or institution that commissions art; **perspective**, the mathematical system for representing depth on a flat surface; **Manueline**, the Portuguese late-Gothic architectural style of the age of discovery.

6

PART TWO - CHAPTER 6

The Age of Discovery

Caravels, compasses and the first global system

THE QUESTION THIS CHAPTER ANSWERS

How did a handful of Atlantic kingdoms join the oceans, and what did that do to the world?

IN THIS CHAPTER, YOU WILL

- √ Explain the motives and technology behind European exploration.
- √ Describe Portugal's route to India and its trading empire.
- √ Explain the Columbian Exchange and its consequences.
- √ Explain why the 1500s count as globalisation's beginning.

THEM

WATCH, READ AND LISTEN

Encyclopaedia Britannica, the Age of Discovery. How European voyages connected the oceans after 1418.

www.britannica.com/event/Age-of-Discovery

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

6.1 Why Europe went to sea

In the fifteenth century, Europeans set out to sail around a continent, and the reasons were strictly practical. Land routes from Asia, along which spices, silk and gold had travelled for centuries, passed through territories controlled by the Ottoman Empire after the fall of Constantinople in 1453, and each middleman raised the price. The Mediterranean powers of Venice and Genoa already dominated the eastern trade. The Atlantic powers, Portugal and Castile, needed a way round the monopoly.

There were also reasons of faith. Both kingdoms had centuries of frontier warfare against Muslim states behind them, and crusading energy survived in a new maritime form. There was a third motive: the European economy needed gold and silver for coinage, and West African gold was already reaching Portugal by trade.

Europe could do this because of a cluster of ship-technology changes: the lateen sail, which let a ship sail into the wind, the caravel hull, light enough to explore rivers and coasts, and, decisively, the magnetic compass and the astrolabe, which together told a captain where he was when the land was out of sight.

YOURTURN

1. Give three motives for European overseas exploration.
2. Explain what the fall of Constantinople did to the spice trade.
3. Why was the caravel better for exploration than older merchant ships?

6.2 Portugal's African route

Portugal moved first, and moved methodically. Year by year from the 1420s, captains pushed further down the West African coast, each voyage commissioned by Prince Henry the Navigator and each one returning with better charts. By the 1440s the Portuguese had reached the gold and slave coasts; in 1482 Diogo Cão planted a stone pillar at the Congo estuary; in 1488 Bartolomeu Dias rounded the Cape of Good Hope and proved the Atlantic and Indian Oceans were joined.

In 1497 **Vasco da Gama** sailed from Lisbon with four ships, followed the coast, swung far into the South Atlantic to catch the winds, and after ten months at sea reached Calicut in India. He returned with a cargo that sold for many times the cost of the expedition. The sea route to India was open, and Lisbon became the entrepôt of Europe.

Portugal then built the first maritime empire in European history: forts and trading posts at Kilwa, Goa, Malacca, Hormuz and Macao. By 1513 the first Portuguese ships reached China; by 1543 they reached Japan. The **Treaty of Tordesillas** of 1494, negotiated with Castile through the Pope, drew a line down the Atlantic and gave Portugal the eastern hemisphere, which is why Brazil speaks Portuguese today.

Date	Event
1415	Portugal takes Ceuta in North Africa
1488	Dias rounds the Cape of Good Hope
1494	Treaty of Tordesillas divides the oceans
1498	Vasco da Gama reaches Calicut
1500	Pedro Álvares Cabral claims Brazil
1513	Portuguese ships reach China

Table 6.1 Milestones of the Portuguese route east.

YOURTURN

1. Explain why Dias's voyage mattered even though he never reached India.
2. What did da Gama's profit on his return cargo prove?
3. How does the Treaty of Tordesillas explain the language of Brazil today?

6.3 Columbus and the collision of two worlds

While Portugal went east, Castile gambled west. **Christopher Columbus**, a Genoese navigator in Spanish service, argued that Asia could be reached by sailing west across the Atlantic. He was wrong about the size of the Earth, but in 1492 he found land: the Americas, though he refused to accept it was not Asia.

The consequences were enormous, and historians call them the **Columbian Exchange**. Maize, potatoes, tomatoes, chillies, cocoa and tobacco crossed to the Old World, transforming diets from Ireland to China. Horses, wheat, cattle and sheep crossed to the New World. So did diseases. Populations in the Americas, who had no immunity to smallpox, measles and influenza, fell by most estimates by around ninety per cent within a century. That demographic catastrophe made possible the Spanish conquest of the Aztec and Inca empires by forces of a few hundred men.

For the peoples of the Americas, discovery was catastrophe. For Europeans, it was the opening of a windfall of silver from Potosí and Mexico that financed European wars and trade with Asia. The first global economy began here, in the 1500s, and you will meet it again in chapter 10, because industrialisation had a prehistory.

YOURTURN

1. What was Columbus's plan, and in what was he mistaken?
2. Define the Columbian Exchange and give two examples in each direction.
3. Explain why diseases mattered more than weapons in the Spanish conquests.

6.4 Circumnavigation and the first global world

In 1519 a Portuguese captain in Spanish service, **Ferdinand Magellan**, sailed south past the American continent into the ocean he called the Pacific. Magellan was killed in the Philippines, but in 1522 the eighteen surviving members of his crew brought one of five ships home, the first people to circle the Earth. With that, the map of the world was complete in outline: one connected ocean system, navigable in principle from any port to any other.

Commerce followed the ships. Silver from Spanish America crossed the Pacific to Manila and bought Chinese silk and porcelain, which crossed back to Acapulco. Slaves from Africa worked the sugar of Brazil and the Caribbean. By 1600 the price of pepper in Lisbon responded to events in the Indian Ocean, and the price of silver in China to production in Peru. The world had become a single system, unevenly and brutally, but a system.

Facts to remember

- Magellan's expedition left with five ships and about 270 men; one ship and 18 men returned.
- The Pacific Ocean was named by Magellan because its waters seemed calm, pacific, after the stormy strait.
- Portugal's empire was built on forts and trading posts, not on settlement: Goa, Malacca, Hormuz, Macao.
- The longitude problem, knowing how far east or west you were, remained unsolved until the marine chronometer of the 1760s.

YOURTURN

1. Why is 1522 a better date than 1519 for the circumnavigation?
2. Explain what connected the silver of Potosí to the silk of China.
3. Why can the late 1500s be called the beginning of globalisation?

6.5 Portugal in the global story: the empire that ran on latitude

Portugal's empire outlasted every rival for five centuries, and its secret was cartographic. The *roteiros*, route books of coastlines, soundings and landmarks, and the *padrão* pillars that Portuguese captains planted on every headland, turned the ocean into a known, repeatable set of sailing instructions. Da Gama's route was hard to find once, and then easy to find forever.

UNIT 6 -

That empire was small in manpower and ran on position: the Cape, Hormuz, Goa, Malacca controlled the choke points of the eastern trade like locks on a canal. When Dutch and English navies arrived in the 1600s with better guns and bigger balance sheets, Portugal kept Brazil, Africa and Macao, and you will meet that survival again in later chapters.

WORDSFORTHISCHAPTER

caravel, a light Portuguese sailing ship suited to exploration; **astrolabe**, an instrument for measuring the height of the Sun or a star above the horizon; **Treaty of Tordesillas**, the 1494 agreement dividing newly found lands between Portugal and Castile; **Columbian Exchange**, the transfer of plants, animals, people and diseases between the hemispheres after 1492; **entrepôt**, a trading post where goods are collected for onward sale.

7

PART TWO · CHAPTER 7

The Reformation

Luther, Henry VIII and the Elizabethan settlement

THE QUESTION THIS CHAPTER ANSWERS

What happened when one Church became many, and kings became popes in their own kingdoms?

IN THIS CHAPTER, YOU WILL

- √ Explain the abuses and the ideas behind the Reformation.
- √ Describe Luther's break with Rome and its spread.
- √ Explain Henry VIII's motives and the English settlement.
- √ Describe the Elizabethan Age and its culture.

THEM

WATCH, READ AND LISTEN

Khan Academy, the Protestant Reformation. From the Ninety-five Theses to the wars of religion.

www.khanacademy.org/humanities/renaissance-reformation

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

7.1 Reform before the storm

In 1500 the Roman Catholic Church was the single largest institution in Europe. It ran universities, hospitals and schools, kept the calendar, and claimed authority over every Christian soul. It had been criticised before, often sharply, and it had survived.

What was new in the early 1500s was the combination of the printing press and a genuinely explosive accusation. The Church sold **indulgences**, certificates that reduced the time a soul spent in purgatory. In 1517 the papacy authorised a new indulgence campaign, partly to fund the rebuilding of St Peter's Basilica in Rome. The thunder it met in Germany changed the world.

The resentment was fed by ordinary grievances: priests who barely understood Latin, fees for every service, church wealth in poor regions. A reform of the head and members of the Church had been demanded for two centuries, by councils, by kings and by reformers such as Erasmus of Rotterdam, who wanted reform without schism, from inside.

YOURTURN

1. What was an indulgence, and what was the money of 1517 meant to fund?
2. Why does printing matter to the story that follows?
3. How did Erasmus's approach differ from what Luther would do?

7.2 Luther and the break

In October 1517, **Martin Luther**, an Augustinian friar and professor of theology at Wittenberg, set out for debate his *Ninety-five Theses*. Whether or not he nailed them to a church door, they were printed, and within months they were read across Germany. His core claim was radical: salvation came by faith alone, through the grace of God, and no sacrament, payment or human institution could buy it.

Luther was excommunicated in 1521 and summoned to the Diet of Worms, where he refused to recant. Protected by his local ruler, Frederick the Wise, he translated the Bible into German, and his translation did for German what the King James Bible would later do for English. Because every believer was to read scripture alone, printing presses, schools and literacy followed the new faith everywhere it went.

The movement fractured Europe along lines rulers could exploit. Princes who broke with Rome stopped the flow of church taxes to Italy and confiscated monastic lands. Lutheranism spread through northern Germany and Scandinavia; Ulrich Zwingli and later **John Calvin**, whose severe, disciplined church took root in Geneva, Switzerland, Scotland and the Netherlands, built other variants.

The Catholic Church answered with its own reform, the **Counter-Reformation**: the Council of Trent clarified doctrine, the Jesuits were founded to teach and to mission, and the Inquisition was strengthened. Europe entered a century of religious war.

YOURTURN

1 State Luther's core claim about salvation.

2 Why did German princes welcome Luther's ideas?

3 Name two things the Counter-Reformation did in response.

7.3 Henry VIII and the break with Rome

England's break with Rome began as a dynastic problem. **Henry VIII** had married Catherine of Aragon, his brother's widow, and by 1527 she had borne no surviving son. Henry wanted the marriage annulled and wanted Anne Boleyn. Pope Clement VII, a prisoner in effect of Catherine's nephew, the Emperor Charles V, could not grant it.

Henry's solution was jurisdictional. A series of acts of Parliament between 1532 and 1534 cut every thread of papal authority in England: the clergy submitted to the king, appeals to Rome were forbidden, and the **Act of Supremacy** of 1534 declared Henry the Supreme Head on Earth of the Church of England. The annulment followed; the new marriage produced not a son but a daughter, Elizabeth. Thomas More, Henry's former chancellor, was executed in 1535 for refusing the oath of supremacy.

Between 1536 and 1541 Henry dissolved the monasteries, some eight hundred religious houses, and sold or granted their lands to nobles and gentry. That single transfer of property gave a large part of the English ruling class a permanent material interest in the break with Rome: it could never be fully undone.

Henry remained theologically conservative, and England's doctrine wobbled under his children: a radical Protestant swing under Edward VI, a Catholic restoration under Mary I, with almost three hundred Protestants burned at the stake, and finally the lasting Protestant settlement of Elizabeth I.

Date	Event in the break with Rome
1527	Henry seeks annulment of his marriage to Catherine of Aragon
1533	Anne Boleyn crowned; Henry excommunicated
1534	Act of Supremacy: Henry Supreme Head of the Church of England
1535	Execution of Thomas More
1536-41	Dissolution of the monasteries
1547-53	Protestant reign of Edward VI
1553-58	Catholic restoration under Mary I

Table 7.1 The English Reformation, 1527 to 1558.

YOURTURN

1. Explain the dynastic problem that started the English Reformation.
2. What did the Act of Supremacy declare?
3. Why did the dissolution of the monasteries make the break hard to reverse?
4. Describe the religious swings under Henry's three children.

7.4 Queen Elizabeth I and the Elizabethan Age

Elizabeth I came to the throne in 1558 and ruled for forty-five years. Her achievement was to settle the religious question her father had opened, by force of moderation. The **Elizabethan Religious Settlement** of 1559 made England Protestant, with the monarch its Supreme Governor, but kept bishops, vestments and a liturgy that Catholics could almost recognise. She is said to have wanted no windows into men's souls: outward conformity, not inward belief, was what the state required.

The second threat was foreign. Catholic Europe regarded her as illegitimate, and in 1588 Philip II of Spain sent the **Armada**, a fleet of about 130 ships, to invade England and restore Catholicism. The English navy, with faster, more heavily gunned ships, and Atlantic storms, destroyed or scattered it. The victory became a founding legend of Protestant England.

The age carries her name for its culture as much as its politics. It is the age of **William Shakespeare**, whose company played at the Globe on the Bankside of the Thames; of Christopher Marlowe; of the composer William Byrd; of the explorers Francis Drake and Walter Ralegh. English merchant companies began their long global reach, and the language itself was in one of its most inventive phases: scholars count that Shakespeare alone first recorded some seventeen hundred words.

Above all, the Elizabethan settlement made England a national state with a national church, and the pattern of authority, Parliament validating each step, set the constitutional lines that the next two centuries of English history would follow.

The Elizabethan Age in numbers

- Elizabeth reigned from 1558 to 1603, the longest reign of any English monarch to that date.
- The Armada of 1588: about 130 ships, roughly half never returned to Spain.
- The Globe Theatre opened in 1599; a full house held about three thousand people.
- Shakespeare wrote around thirty-nine plays and one hundred and fifty-four sonnets.

YOURTURN

1. How did the Religious Settlement balance Protestant doctrine with moderation?
2. Why did Philip II launch the Armada, and why did it fail?
3. Give three features of the Elizabethan cultural flowering.
4. Explain why the constitutional method of the break with Rome mattered later.

7.5 Portugal in the global story: Inquisition, Jesuits and the cycle of faith

Portugal experienced the Reformation era as an anchor rather than a break. The Inquisition was formally established in Portugal in 1536, prosecuting converted Jews and other suspected heretics, and in 1497 the crown had already ordered the forced conversion or expulsion of the Jews, a loss of banking, medical and craft skills that the kingdom never fully repaired.

The Counter-Reformation ran strongly through Portuguese education. The Jesuits founded Saint Anthony's College in Lisbon and, above all, the University of Coimbra regained its European standing, while the Collegio das Artes trained a generation of scholars. At the same time Jesuit missions carried Catholicism to Brazil, Goa, Japan and the Congo, so that the Portuguese seaborne empire became the vehicle of the Counter-Reformation worldwide.

In 1580, when the young King Sebastião died at the battle of Alcácer-Quibir in Morocco in 1578, the Portuguese royal line failed and Philip II of Spain took the throne. For sixty years Portugal and its empire were ruled from Madrid: the *União Ibérica*, which ended in a successful restoration revolt in 1640.

YOURTURN

1 What did Portugal lose through the expulsion and forced conversion of its Jews?

2 How was the Portuguese empire a vehicle of the Counter-Reformation?

3 Explain how Portugal came to be ruled by the king of Spain between 1580 and 1640.

WORDSFORTHISCHAPTER

indulgence, a certificate reducing punishment for sin, sold by the Church before the Reformation; **excommunicate**, to expel someone from the Catholic Church; **Act of Supremacy**, the 1534 act making the English monarch head of the Church of England; **dissolution of the monasteries**, Henry VIII's closure and seizure of religious houses, 1536-41; **Armada**, the Spanish invasion fleet defeated in 1588; **União Ibérica**, the 1580-1640 period of Spanish rule over Portugal.

8

PART TWO – CHAPTER 8

The French Revolution: the old order

The Ancien Regime, the Enlightenment and the causes of revolution

THE QUESTION THIS CHAPTER ANSWERS

Why did the richest state in Europe collapse in a single summer?

IN THIS CHAPTER, YOU WILL

- √ Describe the three Estates and the privileges of each.
- √ Explain Enlightenment ideas about power and rights.
- √ Argue the relative weight of the causes of revolution.
- √ Describe the events of May to August 1789.

THEM

WATCH, READ AND LISTEN

Khan Academy, the French Revolution. From the Ancien Regime to the storming of the Bastille.

www.khanacademy.org/humanities/world-history

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

8.1 The Ancien Régime

France in the 1780s was the richest and most powerful state in Europe, and it was also one of the most rigid. Historians call its system the **Ancien Régime**, the old order. Its structure was a pyramid of privilege agreed by custom and law.

Society was divided into three **Estates**. The First Estate, the clergy, numbered about one hundred thousand people and paid almost no tax, yet collected the tithe, a tenth of every harvest. The Second Estate, the nobility, perhaps four hundred thousand people, held the highest posts in army, church and state, and was largely exempt from the principal direct tax, the *taille*. The Third Estate was everyone else: some twenty-seven million peasants, city workers, and the bourgeoisie, lawyers, merchants, manufacturers, who paid nearly everything and were eligible for nearly nothing.

Government was absolute in theory. The king, Louis XVI, answered, he said, to God alone. In practice the crown was restrained by privilege, by regional variation, by the *parlements*, law courts that registered, and could delay, royal edicts, and above all by debt. The court's home at Versailles consumed a large share of state revenue in ceremony and display, and that visible extravagance would cost the monarchy its remaining goodwill.

Estate	Who	Approximate share of the population	Tax burden
First	Clergy	under 1 per cent	almost none; collected the tithe
Second	Nobility	about 1.5 per cent	largely exempt
Third	Everyone else	about 98 per cent	nearly all direct and indirect taxes

Table 8.1 The three Estates of the Ancien Régime.

YOURTURN

1 Define the Ancien Régime.

2 Who belonged to each of the three Estates?

3 Why is it fair to say the Third Estate paid for the First and the Second?

8.2 Enlightenment ideas

The revolution had an argument before it had an army. The thinkers of the eighteenth-century **Enlightenment** attacked the old order at its foundations. Voltaire mocked the alliance of altar and throne. Montesquieu proposed the separation of legislative, executive and judicial powers as the defence against tyranny. Rousseau argued in *The Social Contract* of 1762 that legitimate authority flows from the people, the **general will**, not from God through the king.

UNIT 8 -

These books were banned in France and read everywhere in France, by the very bourgeoisie that filled the Third Estate. When revolutionaries later wrote the **Declaration of the Rights of Man and of the Citizen** in 1789, its language, rights, liberty, property, resistance to oppression, the general will, came straight from these authors.

The American Revolution of 1776 had already shown that Enlightenment ideas could govern a real state. French officers who fought in America as allies of the new republic, among them the young Marquis de Lafayette, came home with both ideas and ambitions.

> **YOURTURN**
>
> 1. What did Montesquieu propose, and against what danger?
> 2. Explain Rousseau's idea of the general will.
> 3. How did the American Revolution influence France?

8.3 Major causes of the French Revolution

Historians usually group the causes in three baskets, long-term, medium-term and immediate, and you should be able to argue with all three.

1. Long-term: a society built on privilege

The structure of the Estates, described above, was old, widely resented and visibly unjust. Rising literacy and a large bourgeois class with money but no status made reform not just desirable but, in the end, unavoidable.

2. Medium-term: bankruptcy

France was broke. Wars, above all the Seven Years' War of 1756-63, which France lost, and its expensive support for the American Revolution from 1778, had multiplied the debt. By the late 1780s roughly half of all state spending went on servicing it. Attempts at reform failed because every group with privilege defended it: the nobility refused a general land tax and the parlements blocked the edicts. The state could neither tax fairly nor borrow any more.

3. Immediate: harvest failure

In 1788 a hailstorm destroyed the harvest, and the winter that followed was savage. The price of bread, which a Parisian worker's family might spend half its wages on, doubled. In the countryside, villagers faced hunger and, in the old feudal fears, roaming bands of brigands. Spring 1789 was hungry and frightened, and hungry, frightened people do not defer.

Let's break this down

France's social structure had become indefensible: privilege by birth, taxation without consent.

The Enlightenment supplied a vocabulary of rights, sovereignty and citizenship.

Bankruptcy forced the king to summon the Estates-General for the first time since 1614.

Harvest failure in 1788-89 turned structural crisis into immediate desperation.

YOURTURN

1. Explain how war made the French Revolution more likely.
2. Why did each attempt at tax reform before 1789 fail?
3. Which cause do you judge most important, and how would you defend the choice?

8.4 1789: from Estates-General to National Assembly

With no alternative, Louis XVI summoned the **Estates-General** to meet at Versailles in May 1789. Each Estate had equal voting strength, one vote each, so the two privileged orders could always outvote the Third. The Third Estate, whose delegates were overwhelmingly lawyers and educated commoners, demanded voting by head, which would have given them the majority.

In June 1789 the deadlock broke dramatically. The Third Estate declared itself, on Sieyès's argument that it *was* the nation, a **National Assembly**, and swore the Tennis Court Oath not to disperse until France had a constitution. On 14 July, as the king gathered troops, the people of Paris stormed the Bastille, a royal fortress and prison, in search of gunpowder. In August, the Assembly abolished feudal privilege in a single night, and issued the Declaration of the Rights of Man. That story is chapter 9's opening.

YOURTURN

1. What was the voting dispute that deadlocked the Estates-General?
2. Explain the claim behind the National Assembly's name.
3. What was sworn at the Tennis Court Oath?

WORDSFORTHISCHAPTER

Ancien Régime, the political and social order of France before 1789; **Estates-General**, the assembly of the three Estates, rarely summoned; **taille**, the principal direct tax of the Ancien Régime, largely falling on the Third Estate; **Enlightenment**, the eighteenth-century intellectual movement of reason, rights and reform; **general will**, Rousseau's term for the collective interest of the people as the source of authority; **National Assembly**, the body the Third Estate declared itself in June 1789.

9

PART TWO - CHAPTER 9

Revolution, Terror and Napoleon

The Bastille, the Republic, the Terror and the man who settled it

THE QUESTION THIS CHAPTER ANSWERS

How did a revolution for liberty pass through terror and end with an emperor?

IN THIS CHAPTER, YOU WILL

- √ Explain the significance of 14 July 1789.
- √ Describe the shift from monarchy to republic to Terror.
- √ Assess Napoleon as both heir and destroyer of the revolution.
- √ Explain how the Napoleonic wars reached Portugal and Brazil.

THEM

WATCH, READ AND LISTEN

Encyclopaedia Britannica, the French Revolution. The full narrative from 1787 to 1799.

www.britannica.com/event/French-Revolution

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

9.1 The storming of the Bastille, 14 July 1789

The Bastille was a medieval fortress and state prison in eastern Paris whose towers commanded the artisan districts. By July 1789 it held seven prisoners, but it held something more important: gunpowder, and in the popular mind, the entire symbolism of arbitrary royal power.

The immediate cause was fear. Louis XVI had dismissed reforming ministers and massed loyal regiments around Paris. Crowbs believed an attack on the Assembly and on the people was coming. On 14 July a Parisian crowd, joined by mutinous soldiers with cannon, surrounded the Bastille and demanded its surrender. When negotiations failed, they stormed it; about ninety attackers died. The governor's head was carried on a pike through the streets.

The consequences outran the event. The king withdrew his troops. Paris armed itself and formed a city militia that became the National Guard under Lafayette. In the countryside, the Great Fear, a wave of panic about brigands and aristocrats, drove peasants to burn châteaux and destroy feudal records. On the night of 4 August the Assembly abolished feudal privilege in a session of competitive renunciation. The old order was dead in law within three weeks.

14 July is France's national day. It commemorates not a victory over a foreign enemy but a people's seizure of a symbol of their own subjection.

YOURTURN

1 Why was the Bastille attacked if it held almost no prisoners?

2 Explain the connection between the Great Fear and the night of 4 August.

3 Why is it significant that France's national day celebrates an internal event?

9.2 Constitution, war and the fall of the monarchy

The Constitution of 1791 made France a constitutional monarchy with an elected Legislative Assembly, a broad, though not universal, male franchise, and the rights of the Declaration guaranteed in law. It was a moderate settlement, and it lasted barely a year.

Two forces broke it. At home, the king had accepted the constitution in the hope of overturning it; in June 1791 his family's flight to the frontier, the flight to Varennes, where they were recognised and returned under guard, destroyed his credibility. Abroad, the émigrés and European monarchies threatened intervention, and in April 1792 France, its government now dominated by republicans, declared war on Austria and Prussia.

War radicalised everything. In the summer of 1792, as Prussian armies advanced and the fortress commander's threat to shell Paris circulated, Parisians rose on 10 August, took the Tuileries palace, and suspended the king. In September, as the Prussian advance was halted at Valmy, a new body, the **National Convention**, elected by universal male suffrage, met and proclaimed France a **republic**. Louis XVI was tried for treason and guillotined on 21 January 1793.

YOURTURN

1 How did the flight to Varennes damage the monarchy?

2 Explain how foreign war made French politics more radical.

3 What did the Convention proclaim, and on what franchise was it elected?

9.3 The Reign of Terror

By the spring of 1793 the republic was at war with most of Europe, and at war with itself: royalist revolt in the Vendée, federalist revolt in the ports, inflation and hunger in the cities. The Convention's answer was the Committee of Public Safety, dominated from July by **Maximilien Robespierre**, and a policy of revolutionary government.

The **Reign of Terror** ran from September 1793 to July 1794. Its instrument was the Revolutionary Tribunal and the guillotine; its law was the Law of Suspects, under which anyone could be denounced for insufficient revolutionary zeal. Historians count around seventeen thousand official executions, with perhaps a quarter of a million further deaths in the Vendée civil war. The victims were not mainly aristocrats: nobles and clergy made up under a fifth of those guillotined. The majority were peasants and workers accused of counter-revolution, hoarding or desertion.

The Terror had defenders then and now. Its leaders argued, with the foreign armies still on French soil, that the republic could not indulge legal niceties: terror was, in Robespierre's phrase, nothing other than prompt, severe, inflexible justice, and it saved the revolution. Its critics answered then and now that a revolution that eats its defenders, Danton among them, has become the thing it fought.

It consumed itself. In July 1794, the Convention, fearing it was next, turned on Robespierre and guillotined him without trial. The Terror ended the next day, and the reaction set in.

Phase	Dates	Governing body	Character
Constitutional monarchy	1789-92	National / Legislative Assembly	Moderate reform
Republic	1792-93	National Convention	War and radicalisation
Terror	1793-94	Committee of Public Safety	Emergency dictatorship
Directory	1795-99	Five directors	Corrupt instability

Table 9.1 The phases of the French Revolution.

YOURTURN

1 Define the Reign of Terror and give its dates.

2 Why were most victims of the Terror not aristocrats?

3 Present the case for and the case against revolutionary terror.

9.4 Napoleon Bonaparte

Out of the Directory's corruption and the army's victories rose the century's most famous man. **Napoleon Bonaparte**, born in 1769 on Corsica to a minor noble family, was a trained artillery officer of exceptional talent who made his name driving the British from Toulon in 1793 and, in 1796-97, smashing Austrian armies in Italy. In 1799 he returned from an expedition to Egypt and, in the coup of 18 Brumaire, replaced the Directory with a consulate under himself.

He then did what the revolutionaries had failed to do: he made the settlement work. The **Code Civil** of 1804, the Napoleonic Code, put one clear law, with equality before it, over every citizen; it guaranteed the gains of the revolution, careers open to talent, the sale of confiscated lands, secular marriage, and rolled back the position of women. The Concordat of 1801 made peace with the Catholic Church without restoring its power. The Bank of France, the lycées, the prefects, the Legion of Honour: the administrative skeleton Napoleon built still carries France today.

In 1804 he crowned himself Emperor. Thereafter his story is one of empire and overreach: the defeat of Austria at Austerlitz in 1805, the Peninsular War in Spain from 1808, the catastrophic invasion of Russia in 1812 with about six hundred thousand men, of whom perhaps a hundred thousand returned, and finally the defeats of 1813-14. He abdicated, returned for the Hundred Days in 1815, and was broken at **Waterloo** on 18 June 1815. He died in exile on Saint Helena in 1821.

Napoleon in numbers

- Napoleon fought some sixty battles and lost fewer than ten.
- The Code Civil of 1804 runs to 2,281 articles and remains the base of civil law in France and dozens of other countries.
- The Grande Armée that entered Russia in 1812 numbered around 600,000; the rearguard that came out was around 100,000.
- Napoleon's last years were spent on Saint Helena, a British island in the South Atlantic, 1,900 kilometres from the nearest mainland.

YOURTURN

1. List three lasting reforms of the Napoleonic regime.
2. In what ways was Napoleon's rule a continuation of the revolution, and in what ways a reversal?
3. Explain why the Russian campaign of 1812 was fatal to the empire.

9.5 Portugal in the global story: the court that crossed an ocean

The revolutionary wars reached Portugal in a way no one imagined. Portugal, allied to Britain by treaty since the fourteenth century, refused Napoleon's Continental Blockade closing European ports to British trade. In 1807 a French army under Junot invaded; the **royal court, some ten to fifteen thousand people, boarded ships escorted by the Royal Navy and transferred the entire capital of the Portuguese empire from Lisbon to Rio de Janeiro**. A European monarchy ruling from the Americas, ruling in America, was without precedent.

The Peninsular War followed, from 1808 to 1814, with the Anglo-Portuguese army under Wellington and a Portuguese army rebuilt on French lines driving the French out over six campaigns. Meanwhile Brazil, elevated to a united kingdom with Portugal in 1815, acquired banks, a printing press, and the habit of governing itself.

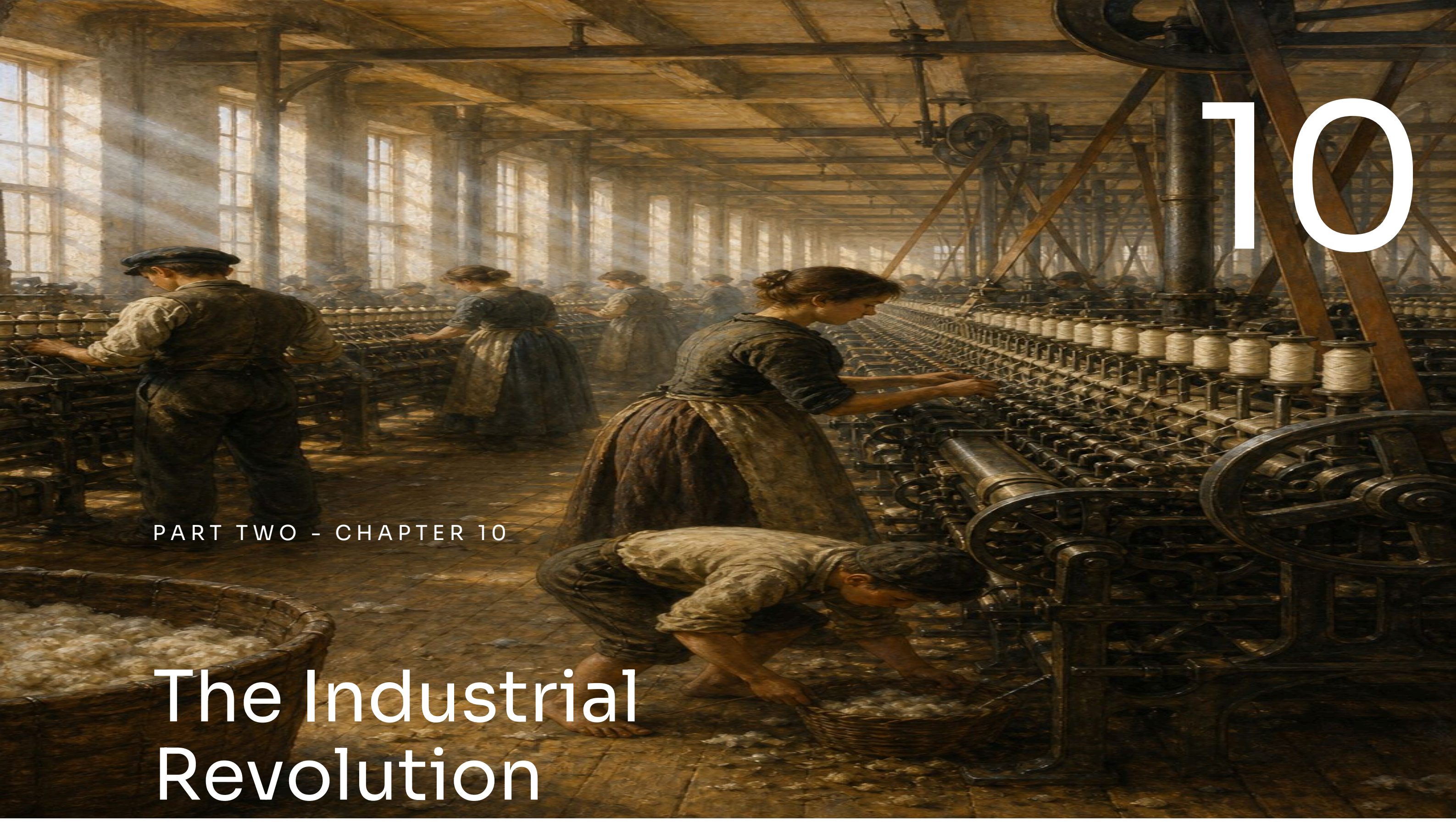
When the court returned in 1821, called by a revolution in Porto, the independence it had accidentally taught was not given up: Brazil declared independence in 1822 under Pedro I, the prince regent himself. The French Revolution's account of rights and nations thus reorganised the Portuguese world without Portugal ever passing through a French-style revolution.

YOURTURN

1. Why did Napoleon invade Portugal, and why did the court go to Brazil?
2. What did Brazil acquire during the thirteen years it hosted the court?
3. Explain how Brazil's independence followed from the court's move.

WORDSFORTHISCHAPTER

Reign of Terror, the period 1793-94 of revolutionary government and mass execution; **Committee of Public Safety**, the emergency executive of 1793-94; **guillotine**, the execution device adopted by the revolution; **Code Civil**, Napoleon's 1804 civil code, the durable settlement of the revolution in law; **Continental Blockade**, Napoleon's closure of European ports to British trade; **Peninsular War**, the 1808-14 war in Spain and Portugal against French rule.



10

PART TWO - CHAPTER 10

The Industrial Revolution

Machines, cities, ideas and the great argument that followed

THE QUESTION THIS CHAPTER ANSWERS

Where and why did the modern economy begin, and what did it cost the people who built it?

THEM

IN THIS CHAPTER, YOU WILL

- √ Explain why industrialisation began in Britain.
- √ Describe working and living conditions in industrial cities.
- √ Explain the rise of unions, socialism and communism.
- √ Compare capitalism and socialism as economic and political systems.

WATCH, READ AND LISTEN

Our World in Data, economic growth. How little output grew before industry, and how fast after.

`ourworldindata.org/economic-growth`

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

10.1 The Industrial Revolution

The Industrial Revolution was the time when the making of goods moved from small workshops and homes to large factories. That single shift changed the culture as well as the economy, because people moved from rural areas to big cities in order to work. It introduced new technologies, new types of transport, and a different way of life for millions.

10.2 Where and why did the Industrial Revolution begin?

The Industrial Revolution began in **Great Britain** in the late 1700s. Many of the first innovations came in the textile industry: the making of cloth moved out of homes, where spinning and weaving had been done by hand, and into large factories. Britain also had plentiful **coal and iron**, the raw materials of power and machinery, and coal seams and iron ore often lay close together and close to ports.

How long did it last?

The Industrial Revolution lasted over a hundred years. Beginning in Britain in the late 1700s, it spread to Europe and the United States. Historians divide it into two phases.

The First Industrial Revolution, from the late 1700s to the mid-1800s, industrialised the manufacture of textiles and moved production from homes to factories. Steam power and the cotton gin played central roles. **The Second Industrial Revolution**, from the mid-1800s to the early 1900s, was the age of mass production in large factories and companies: electricity, the production line and the Bessemer steel process were its key innovations.

When did it start in the United States?

The early American Industrial Revolution took place in the north-east, in New England. Many historians date its start to the opening of **Slater's Mill** in 1793 in Pawtucket, Rhode Island. Samuel Slater had learned textile machinery growing up in England and carried that knowledge to the United States. By the end of the 1800s the United States had become the most industrialised nation in the world.

James Watt's improved **steam engine**, from the 1760s and 1770s, deserves its own sentence. Before Watt, factories had to sit beside rivers, because only water turned a waterwheel. After Watt, a factory could stand anywhere coal could reach, and so could a mine: steam engines pumped water out of deeper and deeper shafts. Energy was cut loose from geography.

YOURTURN

1 Why did the first innovations come in textiles?

2 Distinguish the First from the Second Industrial Revolution.

3 Explain why Watt's steam engine freed factories from riversides.

10.3 Social and cultural changes

Before the revolution most people lived in the country and worked the land. During it they moved to the cities to work in factories. Cities grew, and they grew faster than their housing, drains and water supplies: they became overcrowded, unsanitary and polluted. In many cities poor workers lived in crowded and unsafe buildings. For the average person this was a dramatic shift in the whole shape of a life: where you lived, what you did all day, and who you answered to.

The change in transport was equally dramatic. Where people had travelled by horse, on foot or by boat, the railway, the steamboat and eventually the automobile arrived. Perishable food reached cities from further away; people and products moved around a country, and the world, at speeds that would have seemed magical a lifetime earlier.

YOURTURN

1 Describe the shift in where people lived and worked.

2 How did new transport change ordinary diets?

10.4 Working and housing conditions

One drawback of the Industrial Revolution was the condition of the people inside the factories. There were few laws to protect workers, and the conditions were often dangerous. A generation that had flocked to the cities found that the factory whistle now governed every hour.

Long days

Workers were expected to work long hours or lose their jobs. A twelve-hour day, six days a week, was normal. There was no paid leave. A worker who fell sick or was injured on the job and missed work was often simply fired and replaced.

Dangerous work

Many jobs were dangerous, and there were no government regulations to make them safer. Workers tended powerful machines that had no safety features, and losing a finger or a limb was common enough to be unremarkable. In the mines, men and children worked tunnels that could collapse; below ground, firedamp gas could be ignited by a single spark.

Unsafe facilities

The buildings themselves were hazards. Lighting was poor; dust filled the air of textile mills, damaging lungs over years; and workplaces handling flammable chemicals or fireworks could go up in one blaze.

Child labour

Factories hired children because they were paid less, and, in the smallest operations, because their hands and bodies could reach places adults could not. Children worked the same long weeks under the same conditions as adults, and some were killed or permanently injured. The report of the Sadler Committee of 1832, which took evidence from child workers themselves, shocked the country and made reform unavoidable.

Living conditions

Housing was no better. As more people crowded into the cities, large slums formed, dirty and unsanitary. Whole families sometimes lived in a single room. With people packed so closely and little medical care available, diseases, cholera above all, spread rapidly. Cholera arrived in Britain in 1831-32 and killed tens of thousands before clean water and sewers slowly caught up with the cities.

New government regulations

In the later stages of the Industrial Revolution, workers began to organise into unions to fight for better and safer conditions, and governments, reluctantly, became involved. New regulations shortened the working week and made factories safer. That story, of workers organising and of law slowly following, is topic 10.5's subject.

Facts about working conditions

- In 1860 the five-storey Pemberton Mill in Lawrence, Massachusetts, collapsed, killing an estimated 145 workers; the poorly built factory had been packed to its upper floors with heavy machinery.
- Factories were often very hot in summer and freezing in winter.
- One of the first labour laws was the Factory Act of 1819 in Britain, which made it illegal to employ children under nine.
- As workers organised, they went on strike, refusing to work, to demand better conditions and hours.
- Some early laws actually made it illegal for workers to unionise.

YOURTURN

1 Describe a normal working week for a factory worker in 1840.
2 Why were children employed, and what did that mean for their safety?
3 Why did cholera spread so quickly in industrial cities?
4 What did the Factory Act of 1819 change?

10.5 Labour unions

A **labour union** is a large group of workers, usually in a similar trade or profession, who join together to protect their rights. The Industrial Revolution was the era when national labour unions began to form.

Why did unions first form?

Conditions in factories, mills and mines were terrible, and the government of the day took little interest in safety standards or in how businesses treated their workers. The typical industrial employee worked long hours in danger for little pay. Many were poor immigrants with no alternative. A worker who complained was fired and replaced.

At some point workers began to revolt. They joined together and created unions to fight for safer conditions, better hours and increased wages. The logic was simple and it remains the whole logic of collective bargaining: an owner could easily replace one complaining worker, but not every worker at once, and a factory in which all the workers downed tools earned nothing. One voice is disposable; all the voices together are not.

What did unions do?

Unions organised **strikes** and negotiated with employers for better conditions and pay. During the Industrial Revolution this was rarely a peaceful process. When employers tried to replace striking workers, violence sometimes followed, and occasionally the government had to intervene to restore order, usually on the employers' side.

Major strikes

The **Great Railroad Strike of 1877** began in Martinsburg, West Virginia, after the B&O Railroad cut wages for the third time in a year. It spread across the country; federal troops were sent to put it down; several strikers were killed; and it ended forty-five days after it began. The wages were not restored, but workers had seen the power they held when they acted together.

Other famous strikes included the Homestead Steel Mill Strike of 1892 and the Pullman Strike of 1894. Many ended in violence and the destruction of property, yet over time they began to have a real impact, and conditions gradually improved.

Unions today

Through the 1900s unions became a powerful force in the economy and politics. They are less dominant today, but they still matter in many industries, and the rights they won, the weekend, limits on hours, safe workplaces, are now so normal that it takes an effort to remember they were fought for.

Facts about labour unions

- In 1935 the National Labor Relations Act in the United States guaranteed the right to form a union.
- Business owners sometimes placed spies inside unions and fired any worker who tried to join.
- One of the earliest strikes was held by the Lowell Mill Girls in 1836, who called it a turnout.
- A Chicago strike in 1886 turned into the Haymarket Riot; four strikers were hanged after being found guilty of starting it.

The first of May

May Day, the first of May, has its roots in the late nineteenth century, when workers in industrialised countries began protesting against long hours and poor conditions. A key moment was the Haymarket Affair in Chicago in 1886, when a peaceful demonstration for the eight-hour day turned violent after a bomb was thrown, with deaths and arrests. The tragedy drew worldwide attention to workers' rights, and in 1889 international socialist groups chose the first of May as the day to honour these struggles. It is still a public holiday in Portugal, and in most of the world.

YOURTURN

1. Explain the logic of collective bargaining in your own words.
2. What happened in the Great Railroad Strike of 1877?
3. Why is May Day celebrated on the first of May?

10.6 Socialism and communism

The Industrial Revolution transformed the world faster than almost any period before it. Beginning in Britain around 1760 and spreading across Europe and then the world, it changed how people worked, lived and produced wealth: factories replaced small workshops, machines replaced manual labour, and cities expanded as millions moved from the countryside in search of work.

These changes generated unprecedented economic growth. Production rose dramatically, transport improved, and technological innovation accelerated. But the benefits were not shared equally. Many factory workers faced long hours, dangerous conditions, low wages and poor housing, and entire families, children included, worked to earn enough to survive.

As these problems became impossible to ignore, new political and economic ideas emerged. Among the most influential was **socialism**, which developed during the nineteenth century as a response to the inequalities of early industrial capitalism. Many thinkers, activists and workers concluded that the new industrial economy produced wealth for the owners while leaving the workers in poverty.

Karl Marx and The Communist Manifesto

The most influential figure associated with socialism was **Karl Marx**, born in 1818 in what is now Germany. With Friedrich Engels he published **The Communist Manifesto** in 1848.

The Manifesto was written in a period of rapid industrialisation and social change. Marx and Engels argued that history had been shaped by conflict between social classes, and that industrial society had created a new division: the **bourgeoisie**, who owned the factories and businesses, and the **proletariat**, who worked for wages.

The Manifesto predicted that workers would eventually organise themselves politically, start a revolution, and create a society based on collective rather than private ownership of the major industries. Many of Marx's predictions remain debated by historians and economists, but his ideas had a profound influence on political movements across the world.

It is important to note that socialism existed before Marx and has continued to develop in many forms since. Not all socialists agreed with him, and not all socialist movements aimed at revolution: some believed change could be achieved gradually, through democratic reforms, better labour laws, expanded voting rights and stronger protections for workers.

Let's break this down

The Industrial Revolution created enormous wealth.

Much of that wealth was controlled by factory owners and investors.

Many workers experienced difficult living and working conditions.

Socialist thinkers believed society should distribute wealth more fairly.

Some socialists wanted gradual reform.

Others believed revolutionary change was necessary: those were the communists.

YOURTURN

1. Explain the difference, as the Manifesto sees it, between bourgeoisie and proletariat.
2. What did the Manifesto predict would happen?
3. How did non-revolutionary socialists expect change to come?

10.7 One-party states

During the nineteenth century, workers increasingly organised into trade unions and political movements, campaigning for shorter hours, safer workplaces, better wages and political representation. These movements won reforms now considered normal in modern democracies. But some revolutionary groups, the communists among them, argued that capitalism could never be fully reformed: the system itself created inequality and had to be replaced entirely.

The most significant attempt came in Russia in 1917. The Russian Revolution overthrew the Tsarist monarchy and eventually created the Soviet Union, the first large country to try to organise its economy on socialist principles. Through the twentieth century, socialist and communist governments emerged in China, Cuba, North Korea and several Eastern European states, dividing the world into two major blocs in the period named the **Cold War**.

In practice, many of these governments became **one-party systems**: political systems in which only one party holds power, with opposition parties banned, heavily restricted, or unable to compete effectively.

Supporters of one-party socialist systems argued that political unity was necessary to defend the revolution, prevent social conflict and avoid the return of wealthy elites, and that a single party could represent workers' interests and guide society towards long-term economic goals. Critics answered that one-party systems reduced political freedoms, limited freedom of expression, restricted opposition and concentrated power in a small leadership.

Let's break this down

Socialist revolutions often aimed to create greater equality.

Many revolutionary leaders believed a single party could best achieve that goal.

As a result, several socialist states became one-party systems.

Supporters saw this as necessary for stability and economic planning.

Critics saw it as a limitation on democracy and political freedom.

YOURTURN

1 What was new about the Soviet Union when it was created?

2 Define a one-party system.

3 Summarise the supporters' case and the critics' case.

10.8 Capitalism and socialism compared

To understand modern history, it is useful to compare capitalist and socialist systems both economically and politically.

Capitalism is an economic system in which businesses and industries are owned mainly by private individuals or companies. Economic decisions are shaped by competition, consumer demand and the pursuit of profit. **Communism**, as developed in the Soviet Union and other communist states, aimed at collective ownership of the major means of production: factories, mines, transport networks and other important industries were controlled by the state.

Politically, capitalist countries have often been associated with **liberal democracy**, though capitalism does not automatically require democracy. In a liberal democracy, citizens vote in elections, multiple parties compete for power, and governments are limited by constitutions and laws. Communist states in the twentieth century were usually one-party systems, with limited political competition and greater state control over the economy, the media and public institutions.

In capitalist democracies, citizens generally enjoy broad political freedoms: speech, the press, association. Critics argue, however, that economic inequality can give wealthy individuals and organisations disproportionate influence. In communist systems, governments aimed to reduce inequality and guarantee employment, education and healthcare, but critics argue that political freedoms were far more restricted than in liberal democracies.

Let's break this down

Capitalist systems emphasise private ownership and market competition.

Communist systems emphasise collective ownership and state planning.

Liberal democracies usually allow multiple parties and regular elections.

Communist states often operated under one-party systems.

Capitalist societies may generate high innovation and growth, but also high inequality.

Communist governments aimed at reducing inequality and providing public services, but restricted political freedom.

YOURTURN

1. Define capitalism and communism as economic systems.
2. Why is it wrong to assume capitalism always means democracy?
3. State one strength and one weakness that critics identify in each system.

10.9 Symbols and colours

Certain colours and symbols are commonly associated with these ideologies and help us identify them.

Socialist and communist movements have traditionally been associated with the colour **red**, symbolising workers, solidarity and revolution. Common symbols include the **hammer and sickle**, representing industrial workers and peasants; the **red star**; images of workers standing together; and the **raised fist**, symbolising collective action.

Capitalist societies have no single universal symbol, but they are often represented by images of stock exchanges, banks, private companies, financial districts, factories, entrepreneurs and national currencies. Liberal democratic systems are frequently associated with the **ballot box**, parliamentary buildings, constitutions and symbols of individual rights and political participation.

Symbol	Meaning	Associated with
Hammer and sickle	industrial workers and peasants	communism
Red star	revolution and socialism	communism
Raised fist	collective action and solidarity	socialism, labour movements
Ballot box	elections and political participation	liberal democracy
Red flag	the workers' movement	socialism and communism

Table 10.2 Symbols of the great ideologies.

YOURTURN

1. What does the hammer and sickle represent?
2. Why is red the colour of socialism?
3. What images are associated with liberal democracy, and why?

10.10 Conclusion: the debate that never ended

Understanding the Industrial Revolution together with socialism, communism, capitalism and democracy is to understand most of the major political debates of the modern world. Questions of economic fairness, individual freedom, political participation, workers' rights and the role of government continue to shape societies today, exactly as they did in the nineteenth century when these ideas first appeared in response to industrialisation.

10.11 Capitalism and liberal democracies

To see where these ideas came from, look at how societies have created wealth across history. Economists call the tools, land, resources and systems people use to produce goods and services the **means of production**, and the way wealth was created and distributed has changed with them.

In ancient civilisations, roughly 3000 BCE to 500 CE, wealth was based mainly on land, agriculture and control of labour. In Egypt, Mesopotamia, Greece and Rome the key resource was fertile land, especially near rivers such as the Nile and the Tigris and Euphrates. Large estates were often worked by enslaved people, above all in Rome. Wealth came from owning land and controlling those who worked it, and from trade in grain, metals and textiles; empires added tribute and taxes from conquered territories.

After the fall of the Western Roman Empire in 476 CE, Europe developed **feudalism**, dominant from about the ninth to the fifteenth century. Land remained the source of wealth, but it was organised in a strict hierarchy: kings granted land to nobles, nobles allowed peasants, the serfs, to live and work it, and peasants handed over a portion of their produce and labour. There was very little social mobility: people were born into their position and stayed in it.

From about 1500 to 1750 came **mercantilism**, closely linked to the growth of powerful European states such as Spain, Portugal, France and Britain. Wealth was measured largely by the gold and silver a country possessed. Governments actively controlled the economy, encouraging exports and limiting imports. Colonies supplied raw materials, sugar, tobacco, cotton, and served as markets for finished goods. Trade expanded globally and merchant classes grew in importance.

System	Period	Source of wealth	Key feature
Ancient economies	3000 BCE-500 CE	land, agriculture, labour	slavery and conquest
Feudalism	c. 800-1400	land and peasants	hierarchy with little mobility
Mercantilism	c. 1500-1750	trade and precious metals	state control and colonies
Industrial capitalism	from c. 1760	factories and machinery	private investment and profit

Table 10.3 How the means of production changed.

YOURTURN

1 Define the means of production.

2 What was the basis of wealth in each of the four systems in Table 10.3?

3 Why was social mobility so rare under feudalism?

10.12 The Industrial Revolution and the capitalist system

By the late eighteenth century the Industrial Revolution marked a turning point in the means of production. Instead of relying on land or trade, wealth began to be generated through industrial production: machines powered by water and steam produced goods on a far larger scale, and factories became the centre of economic life as people moved from rural areas to cities to work in them.

This transformation is closely linked to the rise of **capitalism**. In a capitalist system the means of production, factories, machines and resources, are owned by private individuals or companies, whose goal is to produce goods efficiently and make a profit. The Industrial Revolution made capitalism far more powerful by creating new investment opportunities: building factories, railways and machinery required substantial capital, and those who invested successfully could earn significant returns. Wealth was no longer based on land or precious metals but on industrial output, innovation and the ability to compete in the market.

In short: *capitalism is an economic system in which businesses and industries are owned by private individuals rather than the government.*

In a capitalist system

people invest their own money in businesses;

the goal is to make a profit;

prices are decided by supply and demand;

workers are paid wages for their labour.

Why did capitalism grow so quickly during the Industrial Revolution? First, the revolution created investment opportunities: a factory or a railway line cost money but offered the chance of large profits, which attracted investors. Second, production increased massively: machines produced goods far faster than hand labour, so businesses could sell more and earn more. Third, competition became stronger, as rival owners tried to undercut each other, and that competition drove innovation: new machines, better methods, improved efficiency.

There was, however, a downside. Workers often laboured long hours in dangerous conditions for low wages. Capitalism created wealth, and it created inequality.

YOURTURN

1 How did the source of wealth change with industrialisation?

2 List the four features of a capitalist system.

3 Explain the three reasons capitalism grew quickly, and state the cost.

10.13 The Industrial Revolution and liberal democracy

The economic changes had political consequences. As industry expanded, a new **middle class** grew in size and influence: factory owners, merchants and professionals who were not part of the traditional aristocracy but had economic power, and who increasingly demanded political rights. This contributed to the development of liberal democracy, especially in the nineteenth century, after the American and French Revolutions.

Liberal democracy rests on representation, individual rights and the rule of law, and one of its key principles is the protection of **private property**. This became central in an industrial capitalist society: if people were to invest money in factories and businesses, they needed to be sure that what they owned was protected by law. Without that protection there would be little incentive to take risks or invest.

A liberal democracy is a system where

people have the right to vote;

governments are elected;

laws protect individual freedoms;

property is protected by law.

As people gained economic power, they demanded political power as well.

Property rights and registration

Property rights mean that individuals legally own land, buildings and other assets, and that the state recognises and protects that ownership. During and after the Industrial Revolution, governments developed stronger legal systems to guarantee these rights, creating the stability that investment and growth required.

Closely connected is **property registration**. As economies grew more complex, informal agreements were no longer enough to establish who owned what, so official systems were created to record ownership. These made buying and selling safer, disputes easier to resolve and taxation easier to organise, and they reduced the uncertainty on which a functioning capitalist economy cannot operate.

Let's break this down

As societies became more complex, it was no longer enough to say this is mine.

Governments introduced systems to record ownership: property registration.

This allowed clear proof of ownership, safer buying and selling, fewer disputes and better organised taxation.

In many ways this was a foundation of the modern economy.

YOURTURN

1 Who was the new middle class, and what did it demand?

2 Why is property protection essential to investment?

3 What problems does property registration solve?

10.14 Technology, inventions and progress

Strip the Industrial Revolution of its social history and what remains is a list of machines, but the list is the social history. Each invention below changed who could work, where they worked, how much things cost and how fast news travelled, and each one fed the others.

Invention	Inventor	Date	What it changed
Spinning jenny	James Hargreaves	1764	one worker spun many threads at once
Water frame	Richard Arkwright	1769	spinning moved to water-powered factories
Steam engine (improved)	James Watt	1769-76	power anywhere coal could reach
Power loom	Edmund Cartwright	1785	weaving caught up with spinning
Locomotive (Rocket)	George Stephenson	1829	passengers and freight by rail
Telegraph	Samuel Morse	1837-44	messages moved faster than people
Sewing machine	Elias Howe / Isaac Singer	1846-51	clothing became a factory product
Bessemer steel process	Henry Bessemer	1856	cheap steel for rails, bridges, ships
Telephone	Alexander Graham Bell	1876	the human voice down a wire
Practical light bulb	Thomas Edison	1879	the working day and the city after dark
Production line (car)	Henry Ford	1913	mass production at a new scale

Table 10.4 Inventions that changed the world.

Look for the pattern in the table rather than the items. First, **power**: muscle and water gave way to steam, then electricity, each step multiplying the work a single worker could control. Second, **speed**: the locomotive accelerated the movement of goods, and the telegraph, which you will meet again in later chapters, separated information from transport for the first time in history. Third, **scale**: each machine made goods cheaper, which enlarged the market, which justified bigger factories and yet more machines, the self-turning loop that drove the whole age.

The textile machinery of Hargreaves, Arkwright and Cartwright came first, which is why making cloth moved out of homes and into factories, and why the first industrial cities, Manchester above all, were cloth cities. Railways followed the steam engine so closely that within one lifetime a person could cross a country in hours instead of days. Steel then carried the railways, and the industrial city itself, higher and wider. Electricity, at the very end of the sequence, made the factory safe from fire and the working day independent of daylight.

Progress, in the title of this topic, is the loaded word. By every material measure the inventions delivered: goods became cheaper, food travelled further, lifespans lengthened and literacy spread, because printing was mechanised on the same steam power. Yet the mill child of 1840 lived in the middle of the fastest material progress the world had ever seen. The nineteenth century's argument, between the engine's admirers and the engine's victims, between the capitalists who invested and the socialists who organised, is the argument of the whole of this chapter, and it did not end with the century.

Interesting facts about the Industrial Revolution

- Many early factories were powered by water, so they had to sit beside a river that could turn a waterwheel.
- A group of weavers who lost their jobs to large factories fought back by rioting and destroying machinery; they were known as Luddites, after their mythical leader Ned Ludd.
- Printers used steam power to print newspapers and books cheaply, which helped more people get the news and learn to read.
- Some of the most important American inventions of the period were the telegraph, the sewing machine, the telephone, the cotton gin, the practical light bulb and vulcanised rubber.
- Manchester, the centre of the British textile industry, was nicknamed Cottonopolis.

YOUR TURN

1. Name the three patterns that run through Table 10.4, with one example each.
2. Why did textile machinery come first?
3. What did the telegraph do that no invention before it had done?
4. Explain what the word progress hides, using the mill child as your example.

WORDS FOR THIS CHAPTER

socialism, the belief that society should organise the economy for the common benefit rather than private profit alone; **communism**, the revolutionary form of socialism aiming at collective ownership of the means of production; **bourgeoisie**, Marx's term for the owning class; **proletariat**, Marx's term for the wage-earning working class; **liberal democracy**, a system of elected government limited by law, with individual rights; **one-party system**, a political system in which a single party holds power; **means of production**, the tools, land, resources and systems used to produce goods and services; **mercantilism**, the state-directed trading system of about 1500 to 1750; **property rights**, legal ownership protected by the state; **strike**, the collective refusal to work in order to force concessions.

10.15 The Industrial Revolution in Portugal

Britain industrialised in the late 1700s; Portugal did not. For most of the nineteenth century it remained one of the least industrialised countries in western Europe, and the story of why is a lesson in everything this chapter has taught: coal, capital, markets, transport and political stability. Portugal had none of the first four in the quantities Britain had, and the fifth failed twice, in 1807 when the court fled to Brazil, and again in the civil war of 1828 to 1834.

Geography came first. Portugal had almost no coal, and no iron ore to speak of. The rivers that powered early British mills existed, but the market behind them was tiny: a population of about three million, mostly poor, mostly rural, spread along a thin Atlantic strip. Industry needs buyers, and Portugal had few. What capital existed had gone, for two centuries, into trade with Brazil, not into factories at home; when Brazilian gold ran out in the late 1700s the economy had nothing to replace it with.

An early false start. The Marquis of Pombal, the reforming chief minister of King Joseph I, tried to force manufacturing into existence in the 1750s and 1760s: he founded the Royal Silk Manufactory in 1758, protected private glass, woollen and iron works, and created one of Europe's first commercial colleges. But his manufactories were state creations, not market ones, and after his fall in 1777 most declined. The Napoleonic invasions and the loss of Brazil in 1822 finished the job: the empire that had paid for everything was gone.

The Regeneration and the railways. After 1851, under the *Regeneração*, governments bet on what they called material improvement: roads, ports, bridges and above all railways, built largely with British capital and British engineers. The first line, Lisbon to Carregado, opened in 1856; Lisbon and Porto were linked in 1864, and a connection to Spain followed in 1866. The bet was that circulation would create wealth, and it did create the conditions for industry: mills and factories clustered along the lines, above all around Porto, and textiles became the country's leading industry. Wool spinning grew in Covilhã, cotton in Porto and the Ave valley to its north, where water power and later steam drove the looms.

YOUR TURN

1 Explain why the absence of coal and iron mattered so much for Portugal.

2 What was the Regeneration strategy after 1851, and what did it build first?

3 Why did Portuguese industry cluster around Porto and Lisbon rather than spread evenly?

4 Compare the Portuguese experience with Britain's using the conditions from topic 10.2.

Slow growth, deep inequality. Yet industrialisation remained an archipelago: islands of factories in Lisbon, Porto and a few towns, in a sea of subsistence farming. Wages were low, hours were long, and the north's textile valleys became Portugal's own version of the industrial cities of this chapter: crowded, unsanitary and ruled by the factory whistle. The rural south saw almost none of it, and tens of thousands emigrated every decade. Portugal's industrial revolution, when it finally came in the twentieth century, was late, partial and concentrated, and the divide it left between the coastal cities and the interior is still visible today.

Portugal's industrial century in dates

1758 Pombal founds the Royal Silk Manufactory

1856 First railway: Lisbon to Carregado

1864 Lisbon linked to Porto by rail

1866 Rail connection to Spain opens

1890s The textile valleys of the Ave drive the north's industry

1986 Accession to the European Community opens the modern economy

WATCH, READ AND LISTEN

ERIH, the industrial history of Portugal. The European Route of Industrial Heritage tells the whole story, from Pombal's manufactories to the Ave valley mills.

www.erih.net/how-it-started/industrial-history-of-european-countries/portugal

Extension only, and not a substitute for the pages. Ask your teacher before you scan.

Answers

Answers to every closed question in this book.

Chapter 1: The global grid

1.3 1. About 66.5 degrees north and south, which is ninety minus the tilt of 23.5. 2. Inside the Arctic Circle there is at least one day a year when the Sun does not set, and the further north you go the more such days there are. 3. The Arctic is an ocean surrounded by continents; Antarctica is a continent surrounded by ocean. Also: people have lived around the Arctic for thousands of years; nobody lives permanently in Antarctica. 4. Antarctica is a continent covered by an ice sheet kilometres thick with no land able to support settlement or hunting, and it is separated from every other continent by open ocean.

1.4 1. A meridian runs pole to pole and measures longitude; a parallel runs east to west and measures latitude. Meridians are all the same length and meet at the poles; parallels never meet and shrink toward the poles. 2. The Equator is fixed by the shape of the Earth, so latitude has a natural zero; no meridian is different from any other, so a zero has to be agreed. 3. Charts measured from different prime meridians could not be combined, so a position reported by one ship did not mean the same to another. 4. That Paris had a long tradition of astronomical measurement and French charts already used it; it lost because far more of the world's shipping already used Greenwich.

1.5 Discussed in the text; see 2.2 and 2.6 for the time consequences.

Chapter 2: Latitude, longitude and time zones

2.1 1. How far north or south of the Equator a place is, from 0 to 90 degrees. 2. Parallels are circles drawn round a ball and get smaller toward the poles; meridians are all half-circles of the same size. 3. Lower latitude means more direct solar energy and generally warmer. Exception: altitude, so Quito sits on the Equator and is cool because it is high. 4. One is in the northern hemisphere and one in the southern, so their seasons are opposite.

2.2 1. How far east or west of the Prime Meridian, up to 180 degrees. 2. It is the single meridian on the opposite side of the Earth from Greenwich, reached by going 180 degrees either way. 3. Latitude can be read from the height of the Sun or the Pole Star; longitude requires comparing local time with the time at a known meridian, so it needs an accurate clock carried on the voyage. 4. Longitude is measured as a difference in time, so solving one solves the other.

2.3 1. Absolute location is a coordinate pair naming one point; relative location describes a place by its relation to another. 2. Ponta Delgada at 25.7 W and Tokyo at 139.7 E. 3. 16.6 degrees. 4. Because a description depends on the other person already knowing the reference point, and a coordinate does not.

2.4 1. West to east, so the eastern side of you turns to face the Sun first.

moments in different places. 4. A timetable states one time for a departure, and that is meaningless if every station keeps its own clock.

2.5 1. 360 divided by 24 equals 15 degrees per hour. 2. Lisbon keeps UTC in January and Tokyo UTC+9, so 09:00 plus 9 hours gives 18:00. 3. New York is UTC−5 and London UTC, a difference of +5 from New York, so 18:00 plus 5 gives 23:00.

they are bent for convenience; some offsets are half or three-quarters of an hour.

2.6 1. A standard is a reference scale kept by atomic clocks; a zone is what clocks in a place are set to. 2. Atomic clocks keep it accurate, and leap seconds keep it close to the Sun's position over Greenwich. 3. Because mainland Portugal lies close to the Prime Meridian. 4. Because everybody involved uses one unambiguous scale, and because local times change with summer time while UTC does not.

Exactly half. 3. Because the Earth turns, so the Sun is highest at different

Boundaries follow national borders so a country shares one working day, and

2.7 1. Going east clocks run to 12 hours ahead and going west to 12 hours behind, which is 24 hours apart at the same meridian. 2. You skip forward a day.

contain a coordinate pair, and may add a date and a time with its zone.

Chapter 3: Climate zones and biomes

3.1 1. Weather is what the atmosphere is doing now; climate is the long-run pattern; a biome is a large region defined by climate, vegetation and animal life. 2. Latitude, solar energy, temperature, climate, vegetation, biome, human activity. 3. Steep rays concentrate energy on a small area and pass through less atmosphere; shallow rays spread the same energy over more ground. 4. Altitude, distance from the sea, ocean currents, prevailing winds, mountain ranges.

3.2 1. Temperature is constant and high; rainfall varies enormously.

season. The difference is rainfall pattern, not temperature. 3. Wettest March with 362 mm, driest August with 96 mm, annual total 2 694 mm. 4. Nutrients are held in the living plants and heavy rain washes minerals from the soil, so cleared land gives one or two harvests and is then exhausted.

3.3 1. Real seasons: a distinct summer and winter differing in temperature and day length. 2. Temperate forest, western and central Europe; temperate grassland, the Eurasian steppe or North American prairie; Mediterranean, southern Portugal.

possible during a hot dry summer, which is when they are growing.

3.4 1. A region receiving under about 250 mm of rain a year. Rainfall is the word doing the work. 2. Air that rose at the Equator and lost its moisture sinks at about 20 to 30 degrees, and sinking air warms and dries. 3. Distance from any ocean, and rain shadow behind mountains. 4. 13 mm, against Lisbon's 558 mm, so about one fortieth.

3.5 1. Ground that has stayed below freezing for at least two years. Water cannot drain through it, so the thawed surface layer is waterlogged. 2. They cannot draw water through frozen ground, and the growing season is too short and the wind too strong. 3. Yakutsk, because it is far from any ocean, so it has both the colder winter and the warmer summer. 4. The growing season is too short for crops, so food had to come from animals and fish.

3.6 1. Rice, tropical, needs warmth and standing water; wheat, temperate, needs a cool dry ripening season; olives, Mediterranean, need dry summers.

and returned in the other. 4. Acceptance criteria: the rewritten sentence must say that climate sets costs or constraints, and that technology and money can change what those costs allow.

Chapter 4: Hemispheres and global patterns

4.1 1. The northern hemisphere holds most of the land; the southern holds mostly ocean. 2. Because people live on land, and most land is northern.

To keep island groups and countries on a single date. 4. Open answer: it must

Rainforest has heavy rain in every month; savanna has a marked wet and dry

Driest July with 2 mm, annual total 558 mm. 4. To lose as little water as

Steep roofs shed snow, so snow falls there; flat roofs mean it does not.

The monsoon winds reverse twice a year, so ships sailed one way in one season

Because ocean heats and cools far more slowly than land, so a hemisphere that

is mostly ocean has milder extremes. 4. Wholly southern: Australia or Antarctica. Wholly northern: Europe or North America.

4.2 1. The Prime Meridian and the 180th meridian. 2. Europe, Africa, Asia, and Australia. 3. Western, because mainland Portugal lies west of the Prime Meridian. 4. Because the Equator is fixed by the planet's shape and the choice of a prime meridian was an agreement made in 1884.

4.3 1. The western hemisphere is a geographical term set by two meridians; the West is a political and cultural term with no fixed definition. 2. Australia is usually counted in the West and lies far east; Portugal is in the western hemisphere and was a dictatorship inside the western bloc. 3. Because one term describes position and the other describes politics, and no map records politics.

political one, and should date the political claim.

4.4 1. Too dry, such as the Sahara; too cold, such as northern Siberia; too high, such as the Himalaya; too wet or forested, such as the Amazon basin.

is the pupil's, with reasons. 3. Cities founded for a reason that no longer applies still stand there, such as ports founded for sailing ships. 4. Open; credit any answer that names the direction of the error and offers a reason.

Chapter 5: The Renaissance

5.1 1. The rebirth of classical Greek and Roman learning and art between the fourteenth and sixteenth centuries. 2. Wealth from trade, so merchant patrons such as the Medici could pay for art; and physical access to Roman remains and Greek manuscripts. 3. They studied human achievement, classical texts and human dignity rather than only theology.

5.2 1. Parallel lines converge on a single vanishing point, so a flat panel shows depth with mathematical accuracy. 2. To draw weight, muscle and movement correctly; Leonardo's anatomical drawings are also science. 3. Competition between wealthy patrons and cities pushed artists to outdo each other.

5.3 1. Patient observation and mathematics; he lacked a telescope. 2. Because a printed book reached a hundred times more readers than a copied one, so ideas crossed Europe in months. 3. Look, measure, and trust the measurement over old authority.

5.4 1. It applied the same measuring spirit to navigation rather than to painting and sculpture. 2. A Portuguese late-Gothic style whose carvings, ropes, coral, armillary spheres, celebrate the ocean. 3. Its hero is a navigator, not a warrior.

Chapter 6: The Age of Discovery

6.1 1. Bypass the Ottoman-controlled land routes; crusading faith; the need for gold and silver. 2. It put the spice trade under middlemen who raised prices. 3. It was light and manoeuvrable enough to explore coasts and rivers, and with lateen sails it could sail into the wind.

6.2 1. It proved the Atlantic and Indian Oceans were joined, so India was reachable by sea. 2. That the eastern trade was worth investing in repeatedly. 3. The treaty gave Portugal the eastern hemisphere; Brazil fell on Portugal's side of the line.

6.3 1. To reach Asia by sailing west; he thought the Earth smaller than it is. 2. The transfer of plants, animals, people and diseases after 1492; eastwards maize and potatoes, westwards horses and wheat. 3. Diseases killed most of the American population, so resistance collapsed.

6.4 1. Because the voyage was only complete when the survivors returned in 1522. 2. Silver crossed the Pacific to Manila and bought Chinese silk for the American and European markets. 3. Because prices in one continent now responded to events in another.

6.5 1. Roteiros and padrão pillars made ocean routes repeatable, and the empire held the choke points of the eastern trade. Accept any two. 2. Holland and England arrived with more capital and stronger navies; Portugal kept Brazil, Africa and Macao.

Chapter 7: The Reformation

7.1 1. A certificate reducing punishment for sin; the rebuilding of St Peter's Basilica. 2. Because it let accusations travel faster than the Church could answer them. 3. Erasmus wanted reform from inside, without a break.

7.2 1. Salvation by faith alone, through God's grace; no payment or institution could buy it. 2. Ending church taxes to Rome and gaining monastic lands. 3. Any two of: the Council of Trent, the Jesuits, the strengthened Inquisition.

7.3 1. He needed a son; Catherine had none; the Pope, controlled by Charles V, would not annul the marriage. 2. That Henry was Supreme Head on Earth of the Church of England. 3. Because the buyers of monastic land could never allow a full restoration of Rome. 4. Protestant under Edward VI, Catholic under Mary I, the lasting Protestant settlement under Elizabeth I.

7.4 1. Protestant doctrine, but with bishops, vestments and a liturgy designed so that moderation would hold. 2. To invade England and restore Catholicism; faster English ships, better guns, Atlantic storms. 3. Shakespeare, Marlowe, Byrd, Drake, Ralegh; the inventive expansion of English itself. 4. Each step had been validated by Parliament, a precedent for constitutional change.

7.5 1. Banking, medical and craft skills of the Jewish community. 2. Jesuit missions carried Catholicism through the empire to Brazil, Goa, Japan and the Congo. 3. Sebastião died without an heir in 1578, and Philip II of Spain claimed the throne in 1580.

Chapter 8: The French Revolution

8.1 1. The political and social order of France before 1789: monarchy by divine right, three Estates, privilege by birth. 2. First: clergy. Second: nobility. Third: everyone else, from rich lawyers to landless labourers. 3. Because the first two were largely exempt and the third paid almost all direct and indirect taxes.

8.2 1. The separation of legislative, executive and judicial powers, as a defence against tyranny. 2. Legitimate authority comes from the people as a whole, not from God through the king. 3. It showed Enlightenment ideas working in a real state, and French officers came home with the experience.

8.3 1. Wars multiplied the debt until servicing it consumed half of spending, forcing the king to summon the Estates-General. 2. Every privileged group defended its exemptions; the nobility refused a general land tax and the parlements blocked the edicts. 3. Any defended choice; strongest answers link bankruptcy to the immediate trigger.

8.4 1. The Third Estate wanted voting by head, which would give it the majority; the other two wanted one vote per Estate. 2. Sieyès's argument that the Third Estate was the nation, so its assembly was the national one. 3. Not to disperse until France had a constitution.

Chapter 9: Revolution, Terror and Napoleon

9.1 1. Gunpowder, and the symbolism of arbitrary royal power. 2. Rural panic drove peasants to burn châteaux, and the frightened Assembly abolished feudal privilege on 4 August. 3. Because the event is the people seizing a symbol of their own subjection, not a foreign victory.

9.2 1. It showed the king's acceptance of the constitution was false and destroyed trust in him. 2. Defeats and invasion scares pushed politics leftward and made moderation look like treason. 3. A republic, elected by universal male suffrage.

9.3 1. September 1793 to July 1794, revolutionary government by the Committee of Public Safety with mass execution of suspects. 2. Because most suspects were ordinary people denounced for hoarding, desertion or insufficient zeal. 3. For: it saved the republic under invasion. Against: it became the tyranny it fought. Both sides need evidence.

9.4 1. Any three: the Code Civil, the Concordat, the Bank of France, the lycées, the prefects. 2. Continuation: equality before the law, careers open to talent, the lands of émigrés kept by buyers. Reversal: emperor, censorship, the position of women. 3. Six hundred thousand men entered; perhaps a hundred thousand returned; the empire never recovered.

9.5 1. Portugal honoured its British alliance and refused the Continental Blockade; the court escaped by sea to Rio under Royal Navy escort. 2. Banks, a printing press, and the habit of self-government. 3. Thirteen years of hosting the court taught Brazil to govern itself, and Pedro I led independence in 1822.

Chapter 10: The Industrial Revolution

10.1-10.2 1. Because cloth was the largest market, its production was already commercialised, and its machinery was simplest to mechanise. 2. First: textiles, water and steam, 1760s-1850s. Second: electricity, steel, mass production, 1850s-1900s. 3. Because Watt's engine worked anywhere coal could reach, not only beside a river.

10.3 1. From rural villages and farm work to industrial cities and factory work, governed by the clock. 2. Perishable food reached cities from further away, so diets widened.

10.4 1. Twelve hours a day, six days a week, no paid leave, dismissal for sickness. 2. They were cheap and small enough for narrow spaces; their safety was nobody's priority. 3. Crowded single rooms, no clean water, no sewers, little medicine. 4. It made it illegal to employ children under nine.

10.5 1. Accept answers built on: one complaining worker is disposable; all of them together are not. 2. Wage cuts at B&O triggered it; it spread nationally; federal troops ended it after 45 days; wages were not restored but workers saw their power. 3. It marks the Haymarket Affair of 1886; socialist groups chose the date in 1889.

10.6 1. Bourgeoisie: owners of factories and businesses. Proletariat: wage-earning workers. 2. That workers would organise, revolt and create collective ownership. 3. Through democratic reform, labour law and expanded voting rights.

10.7 1. It was the first large state to organise its economy on socialist principles. 2. A system in which one party holds power and opposition is banned or restricted. 3. Supporters: unity, stability, planning. Critics: lost freedoms, concentrated power.

10.8 1. Capitalism: private ownership and profit. Communism: state ownership of the means of production. 2. Because capitalism has coexisted with many non-democratic regimes. 3. Capitalism: innovation and growth, but inequality. Communism: guaranteed services, but restricted freedom.

10.9 1. Industrial workers and peasants. 2. It is the colour of workers' movements and revolution. 3. Ballot boxes, parliaments, constitutions: participation and individual rights.

10.11 1. The tools, land, resources and systems used to produce goods and services. 2. Ancient: land, agriculture, labour. Feudal: land and peasant dues. Mercantile: trade and bullion. Industrial: factories and machinery. 3. Because position was fixed by birth into the hierarchy.

10.12 1. From land and trade to industrial output, innovation and market competition. 2. Private investment, profit, prices by supply and demand, wages for labour. 3. Investment opportunities, machine speed, competitive innovation; the cost was worker exploitation and inequality.

10.13 1. Factory owners, merchants and professionals with money but no aristocratic status. 2. Because without legal protection of ownership no one would risk investment. 3. Proof of ownership, safer transactions, fewer disputes, workable taxation.

10.14 1. Power: muscle to steam to electricity. Speed: the railway, the telegraph. Scale: cheaper goods, bigger markets, bigger factories. 2. Because cloth was the biggest market and its machines came first. 3. It separated information from transport: messages moved faster than people for the first time. 4. The mill child lived amid the fastest material progress in history; averages hide lives.

10.15 1. With no coal it could not use steam power, and with no iron it could not build machines, so early industry had no cheap energy or equipment. 2. Material improvement: first roads, ports, bridges and above all railways, built with British capital. 3. Because factories clustered along the railways and the textile rivers of the north; the rural south had neither markets nor transport. 4. Strong answers test Portugal against coal, iron, capital, labour, markets and transport, and find it lacking in all six.

Glossary

Every word this book teaches, defined as it is used here, with the unit that introduces it.

absolute location A position given as a coordinate pair, naming one point on Earth and no other. Unit 2

Arctic Circle The parallel at about 66.5 degrees north, inside which the Sun does not set on at least one day a year. Unit 1

axial tilt The lean of the Earth's axis, about 23.5 degrees, which does not change through the year and causes the seasons. Unit 1

axis The imaginary line through the Earth about which it turns, tilted at about 23.5 degrees. Unit 1

biodiversity The variety of living things in a place, counted as species or as the range of forms. Unit 3

biome A large region of the Earth defined by its climate, its vegetation and the animals that live in it. Unit 3

chronometer An accurate clock built to keep reference time at sea, which is what made longitude measurable on a voyage. Unit 2

climate The pattern weather makes over a long period, usually thirty years or more. It is described, not forecast. Unit 3

containment The policy of preventing the spread of an opposing system rather than attacking it. Unit continental climate A climate far from the sea, with hot summers and cold winters. Unit 4

Coordinated Universal Time The world time standard, kept by atomic clocks, from which every time zone is described as an offset. Unit 2

density How many of something there are in a given area, such as people per square kilometre. Unit 4

desert A region receiving under about 250 millimetres of rain a year. The definition is about rainfall, not heat. Unit 3

distribution How something, usually population, is spread across an area. Unit 4

Equator The imaginary line half way between the poles, at zero degrees latitude, fixed by the shape of the Earth. Unit 1

grassland A biome of continental interiors where rainfall is too low for forest, and the soils are often very productive. Unit 3

Greenwich Mean Time The time zone kept by clocks in Britain in winter, and the ancestor of the modern world standard. Unit 2

growing season The part of the year in which conditions allow plants to grow. Unit 3

hemisphere Half of the Earth, divided either by the Equator or by a pair of meridians. Unit 1

International Date Line The line, roughly along the 180th meridian, where the calendar date changes. Unit 2

latitude How far north or south of the Equator a place is, measured as an angle from 0 to 90 degrees. Unit 1

longitude How far east or west of the Prime Meridian a place is, measured as an angle up to 180 degrees. Unit 1

maritime climate A climate near the sea, whose extremes are moderated by water. Unit 4

Mediterranean climate A climate with mild wet winters and hot dry summers, in which plants are adapted to drought in the growing season. Unit 3

meridian A line running from the North Pole to the South Pole. All meridians are the same length. Unit 1

midnight Sun A period inside a polar circle when the Sun does not set at all. Unit 1

offset The number of hours a time zone differs from Coordinated Universal Time. Unit 2

parallel A line of equal latitude. Parallels never meet and get smaller toward the poles. Unit 1

permafrost Ground that has stayed below freezing for at least two years, through which water cannot drain. Unit 3

polar night A period inside a polar circle when the Sun does not rise at all. Unit 1

Prime Meridian The meridian at zero degrees longitude, running through Greenwich, agreed internationally in 1884. Unit 1

rain shadow The dry area behind a mountain range, where air has already lost its moisture. Unit 3

rainforest A tropical biome with heavy rain in every month, a closed canopy and the highest biodiversity on land. Unit 3

rotation The turning of the Earth about its axis, once in about twenty-four hours, from west to east. Unit 2

savanna A tropical biome of grassland with scattered trees, produced by a marked wet season and dry season. Unit 3

solar energy The energy arriving from the Sun, which is concentrated where its rays strike steeply and spread where they strike at an angle. Unit 3

taiga The great coniferous forest of the north, the largest forest on Earth. Unit 3

temperate forest A biome of the middle latitudes with rain through the year and trees that often drop their leaves in autumn. Unit 3

time zone A band of longitude within which clocks are set to the same time. Unit 2

Tropic of Cancer The parallel at about 23.5 degrees north, the furthest north the Sun can be directly overhead. Unit 1

Tropic of Capricorn The parallel at about 23.5 degrees south, the furthest south the Sun can be directly overhead. Unit 1

tundra A polar biome of low mosses, lichens and dwarf shrubs above permanently frozen ground. Unit 3

weather What the atmosphere is doing now or this week. It is forecast, and it changes hour by hour. Unit 3

caravel A light Portuguese sailing ship suited to exploration. Unit 6

astrolabe An instrument for measuring the height of the Sun or a star above the horizon. Unit 6

Treaty of Tordesillas The 1494 agreement dividing newly found lands between Portugal and Castile. Unit 6

Columbian Exchange The transfer of plants, animals, people and diseases between the hemispheres after 1492. Unit 6

entrepôt A trading post where goods are collected for onward sale. Unit 6

indulgence A certificate reducing punishment for sin, sold by the Church before the Reformation. Unit 7

excommunicate To expel someone from the Catholic Church. Unit 7

Act of Supremacy The 1534 act making the English monarch head of the Church of England. Unit 7

dissolution of the monasteries Henry VIII's closure and seizure of religious houses, 1536-41. Unit 7

Armada The Spanish invasion fleet defeated in 1588. Unit 7

União Ibérica The 1580-1640 period of Spanish rule over Portugal. Unit 7

Renaissance The rebirth of classical learning and art from the fourteenth to the sixteenth century. Unit 5

humanism The study of human achievement, classical texts and human dignity. Unit 5

patron A wealthy person or institution that commissions art. Unit 5

perspective The mathematical system for representing depth on a flat surface. Unit 5

Manueline The Portuguese late-Gothic architectural style of the age of discovery. Unit 5

Ancien Régime The political and social order of France before 1789. Unit 8

Estates-General The assembly of the three Estates, rarely summoned. Unit 8

taille The principal direct tax of the Ancien Régime, largely falling on the Third Estate. Unit 8

Enlightenment The eighteenth-century intellectual movement of reason, rights and reform. Unit 8

general will Rousseau's term for the collective interest of the people as the source of authority. Unit 8

National Assembly The body the Third Estate declared itself in June 1789. Unit 8

Reign of Terror The period 1793-94 of revolutionary government and mass execution. Unit 9

Committee of Public Safety The emergency executive of 1793-94. Unit 9

guillotine The execution device adopted by the revolution. Unit 9

Code Civil Napoleon's 1804 civil code, the durable settlement of the revolution in law. Unit 9

Continental Blockade Napoleon's closure of European ports to British trade. Unit 9

Peninsular War The 1808-14 war in Spain and Portugal against French rule. Unit 9

socialism The belief that society should organise the economy for the common benefit rather than private profit alone. Unit 10

communism The revolutionary form of socialism aiming at collective ownership of the means of production. Unit 10

bourgeoisie Marx's term for the owning class. Unit 10

proletariat Marx's term for the wage-earning working class. Unit 10

liberal democracy A system of elected government limited by law, with individual rights. Unit 10

one-party system A political system in which a single party holds power. Unit 10

means of production The tools, land, resources and systems used to produce goods and services. Unit 10

mercantilism The state-directed trading system of about 1500 to 1750. Unit 10

property rights Legal ownership protected by the state. Unit 10

strike The collective refusal to work in order to force concessions. Unit 10

labour union An organisation of workers joined to protect their rights by collective action. Unit 10

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